



Synergistic disturbance rejection for tail-sitter UAV hover mode and transition phase: vectoring rotors and bioinspired phase-shifting wings

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ABSTRACT

In recent years, a class of hybrid convertible unmanned aerial vehicles (UAVs) with vertical take-off and landing (VTOL) capability and cruise mode transition capacity has attracted extensive research attention. Tail-sitter UAVs stand out among these UAV platforms for their inherent advantage of no redundant actuation resources. However, tail-sitter UAVs face notable challenges in terms of disturbance rejection performance during both hover mode and transition phase. To address this critical limitation, this paper proposes a synergistic disturbance rejection scheme integrating vectoring rotor mechanisms and phase-shifting wing mechanisms. Accordingly, a novel tail-sitter UAV configuration, designated the phase-shifting wing tail-sitter UAV (PW-TUAV), is developed in this work. Based on multibody dynamics theory, a full dynamic model of the PW-TUAV is established. This model incorporates the coupled aerodynamic characteristics arising from wing translation and rotation, along with the non-minimum-phase characteristics of the vectoring rotors. In parallel, a full-attitude disturbance rejection synergistic controller (SC) on SO(3) space is developed, complemented by an integral-driven dynamic control weight allocation strategy. Comparative simulation experiments with ten sets of random wind disturbance samples were performed to quantify the theoretical performance improvement of the proposed scheme. In hover mode, the proposed scheme delivers a theoretical 114.7% improvement in the UAV's forward wind disturbance rejection capability, alongside a 171.2% enhancement in its local wind disturbance rejection capability. During the transition phase, the proposed scheme further achieves a 105.1% enhancement in the UAV's disturbance rejection performance. An initial prototype of the PW-TUAV is fabricated, and outdoor flight experiments covering VTOL and transition phases are conducted under Level 3 natural wind disturbance. The experimental results validate the effectiveness of the proposed scheme in enhancing the stability and reliability of this class of tail-sitter UAVs with a wingspan of approximately 1 m and a self-weight of around 0.5 kg.

1. Introduction

In recent years, unmanned aerial vehicles (UAVs) have experienced rapid development owing to their broad application value. They have been maturely deployed in numerous fields, including search and rescue [1,2], visual surveillance [3,4], agricultural operations [5,6], and military missions [7]. However, traditional UAVs are constrained by their inherent flight mechanisms, resulting in respective insurmountable technical obstacles [8]. For instance, fixed-wing UAVs face inherent difficulties in takeoff and landing. Multirotor UAVs suffer from inherent drawbacks of low flight efficiency and limited horizontal flight speed. Flapping-wing UAVs are restricted by material strength constraints, which result in a low upper limit of driving power [9]. Against this

backdrop, hybrid and convertible vertical take-off and landing (VTOL) UAVs have attracted considerable attention. Characterized by multiple flight modes, these UAVs can combine the advantages of various types of UAVs, thereby better meeting the increasingly complex and diverse demands of civil and military missions [10].

However, hybrid drive technology remains in the exploratory stage, and no configuration yet perfectly accommodates multiple flight modes. Most designs suffer from underutilized drive resources. Quadplane [11–13] and tandem-wing [14] UAVs are typical representatives of redundant designs. They typically require five or more rotors, and are equipped with two independent control systems for VTOL and cruise mode respectively. This design leads to complete drive redundancy, additional payload weight, and adverse aerodynamic effects. To

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alleviate the above full redundancy issue, tiltrotor UAVs [15–17] are proposed, which achieve flight mode switching through 90° rotation of the rotor system. Nevertheless, they require at least 3–4 symmetrically arranged rotors to maintain balance during the VTOL phase. This structural design leaves part of the drive resources in a redundant state during cruise mode. Lifting-wing UAVs [18,19] are further optimized for drive resource reuse, which integrate fixed wings and rotors at a special installation angle. This design enables the full reuse of drive units without additional traction rotors. However, the narrow small-angle operation range of the rotors directly limits the maximum forward propulsion power of the UAVs.

From the perspective of the utilization efficiency of drive resources, tail-sitter UAVs [20–23] may be the most effective design to date. They use only one powertrain for both horizontal and vertical flight, without any redundant drive resources [24]. Among these designs, the configuration with two rotors [25–28] can further meet lightweight requirements, and its greatly reduced horizontal dimensions also enable it to navigate through narrow spaces. However, at the same time, tail-sitters require the fuselage to perform large-angle maneuvering during the transition phase between different flight modes, which poses enormous challenges to control algorithms. In addition, the vertically mounted wings result in a larger windward area, leading to poor wind disturbance rejection performance during hover mode. Therefore, tail-sitters still cannot be widely applied to actual outdoor flight missions so far.

Several studies have been dedicated to enhancing the anti-disturbance performance of tail-sitters. Tilted-wing tail-sitters [29–31] fold their wings along the spanwise direction, enabling them to converge into quadrotor UAVs during VTOL phase while significantly reducing the windward area, thus achieving stable hovering. However, this design still has two notable limitations. On the one hand, excessive aerodynamic torque generated during high-speed cruise mode imposes strict constraints on the driving torque and hinge mechanism strength required for wing folding and unfolding [31]. On the other hand, the design makes the transition phase of the aircraft more complex. The complex transition phase is difficult to model and control accurately, which further increases the control difficulty of the aircraft during transition phase. Bioinspired wing tail-sitters [32] mimic the wing-folding motion of birds in nature. This bionic design can reduce the windward area of the aircraft during the VTOL phase. However, the overly complex wing structure severely hinders the lightweight design of the aircraft. Moreover, similar schemes only enhance stability for hovering control and fail to provide stability gains for the transition phase, which requires more robust anti-disturbance capability. In fact, for tail-sitters, hover mode is only a short-duration phase for low-altitude takeoff and landing [33]; most crash risks occur during the transition phase affected by gust disturbances. Particularly for tail-sitters equipped with only two rotors to maintain their aerodynamic shape, the effectiveness of aerodynamic control surfaces is limited at low airspeeds. During landing, airflow around the aerodynamic control surfaces decreases or even reverses [34,35], resulting in significant deficiencies in anti-disturbance performance. Tail-sitters integrated with the vectoring rotor configuration [27,28] feature dynamically adjustable propeller thrust directions, which provide sufficient control torque even at low airspeeds, allowing tail-sitters to perform takeoff, landing, and transition phase more stably. Nevertheless, adverse aerodynamic effects still persist. The scheme of maintaining aircraft attitude by increasing control torque is not efficient, and in the face of strong disturbances, it may lead to loss of control due to highly coupled control torques. Exploring a solution that passively maintains balance or enables tail-sitters to adapt to environmental requirements by modifying wing aerodynamics may be a more efficient approach.

The flight mechanisms of birds in nature have not been fully elucidated to this day. They can always achieve highly stable maneuvering flight through subtle yet elegant movements, thanks to their wings that can adjust shapes at will and robust environmental perception

capabilities. Inspired by bionics, many types of morphing-wing UAVs have emerged, but most focus on the study of wing folding and unfolding [36–38] and full-span pitching [39,40]. Several studies on ultra-lightweight flapping-wing UAVs [41–43] have revealed the influence of wing phase angles on aerodynamic characteristics. However, transferring this technology to UAVs with heavier fuselages and higher flight speeds faces challenges posed by limited driving torque and insufficient strength of hinge mechanisms.

In summary, most existing tail-sitter UAVs generally suffer from insufficient disturbance rejection performance in hover mode and transition phase. Meanwhile, existing optimization strategies have inherent limitations in multiple aspects, including configuration reliability, lightweight design, reusability across multiple flight modes, modeling complexity, and control efficiency. Against this background, this paper aims to propose a novel bio-inspired configuration, followed by the establishment of its unique aerodynamic model and the investigation of the corresponding synergistic controller (SC). The proposed scheme features the advantages of a lightweight and reliable structure, full reusability of actuation resources across all flight modes, facile reduced-order dynamic modeling, and straightforward controller design.

The primary contributions of this research include:

- (1) A novel tail-sitter configuration is designed, which is named the phase-shifting wing tail-sitter UAV (PW-TUAV). It features a tailless layout, comprising a pair of vectoring rotors mounted at the aircraft nose and a pair of bioinspired wings with independent morphing capability. A special flapping-wing drive mechanism with self-locking capability is designed, thus reducing the requirement for the output torque capability of servo actuators.
- (2) The multibody modeling approach is adopted to systematically analyze the dynamic characteristics of the proposed new-configuration tail-sitter. This includes the analysis of the complex aerodynamic characteristics between the translational and rotational motions of the flappable wings, as well as the analysis of the non-minimum-phase characteristics of the vectoring rotor mechanism.
- (3) Based on the analysis of the response characteristics of each actuator, an easily implementable integral-driven dynamic control weight allocation strategy is proposed, and a full-attitude disturbance rejection SC for the aircraft is designed in the SO (3) space.
- (4) To further characterize the theoretical performance advantage of the proposed scheme, comparative simulation tests using ten independent sets of random wind disturbance samples were implemented. When operating in hover mode, the proposed scheme brings a theoretical 114.7% enhancement in the UAV's forward wind disturbance rejection capacity, along with a 171.2% improvement in its local wind disturbance rejection ability. In the transition phase, the proposed scheme also delivers a 105.1% boost in the UAV's full-range disturbance rejection performance.
- (5) A PW-TUAV prototype is developed, and a flight test platform is established, with successful completion of hover mode and transition phase flight experiments. Experimental results demonstrate that under random disturbances of natural wind at level 3 or below, the PW-TUAV can resist lateral displacement with a maximum instantaneous horizontal displacement velocity not exceeding 0.5 m/s. The stable transition phase from hover mode to cruise mode can be smoothly executed in approximately 3.3 s. Even subjected to multi-axis coupled disturbances of roll angle and yaw angle up to 15.5° during the transition phase, the aircraft can quickly return to a stable state. These results verify the significance of the proposed scheme in enhancing the stability of tail-sitters, thus broadening their application prospects.

The rest of the article is organized as follows. Section II introduces the system configuration design of the PW-TUAV. Section III derives the multibody dynamics model of the flight platform. Section IV presents the controller design scheme. Section V conducts simulation analysis and prototype testing experiments. Section VI summarizes the conclusions and outlines future work. For quick reference, a full list of acronyms used throughout this paper is compiled in [Appendix A](#), and the definition methodology for all kinematic variables adopted in the kinematic and dynamic derivations is detailed in [Appendix B](#).

2. System configuration design

To realize the aforementioned goals of a lightweight, reliable structure and full reusability of actuation resources across all flight modes, we propose the following PW-TUAV configuration. This section first introduces the structural characteristics of the PW-TUAV configuration and the material properties of its prototype, then elaborates on the unique control mechanism adopted in this design. In addition, optimization is carried out on the drive mechanism of the phase-shifting wing to endow it with self-locking capability, which reduces the performance requirements for micro servos, thus enabling the first application of this technology to tail-sitter UAVs.

2.1. Structural layout and materials

[Fig. 1](#) illustrates the main layout and material composition of the designed PW-TUAV prototype. It consists of four basic modules, namely the airframe module, propulsion module, wing module, and landing module. The main body is entirely made of hollow carbon fiber, while the connecting components are all fabricated from 3D-printed nylon, so as to meet the requirements of both lightweight design and structural strength to the greatest extent possible. Two hollow carbon fiber square tubes with dimensions of $10 \text{ mm} \times 10 \text{ mm}$ are employed as the core support structures for the airframe module and propulsion module respectively, and are firmly connected via 3D-printed angle brackets to ensure the transmission of forces and torques. The wing module is mainly assembled from hollow carbon fiber rods with a diameter of 2–3 mm, using needle and thread for stitching, and then covered with a layer of 0.1 mm-thick P31N tear-resistant fabric. This design results in a single wing weighing only around 17 g, accounting for approximately 3.5% of the total prototype weight. Since the primary objective of this research is to verify the contribution of the phase-shifting wing to the disturbance rejection achieved by the SC rather than to enhance cruise flight efficiency, the airfoil is temporarily designed as a flat wing without camber. This design also enables the aircraft to possess almost identical flight capabilities whether it is in a back-up or belly-up orientation, implying potential for performing special maneuvering actions.

The key design parameters of the PW-TUAV are presented in [Table 1](#). Among these, the main airframe structure refers to the assembly of all

Table 1
PW-TUAV prototype specifications.

Parameters	Symbol	Value
Total mass of prototype [kg]	m_{sys}	0.49
Mass of main airframe structure [kg]	m_b	0.382
Mass of a single wing [kg]	m_w	0.017
Mass of a single vectoring rotor [kg]	m_p	0.037
Moment of inertia of main airframe structure [kg·m ²]	J_b	diag{5.12, 4.03, 1.37} × 10 ⁻³
Moment of inertia of a single wing [kg·m ²]	J_w	diag{44.8, 8.06, 36.8} × 10 ⁻⁵
Moment of inertia of a single vectoring rotor [kg·m ²]	J_p	diag{6.19, 6.10, 7.46} × 10 ⁻⁶
Wingspan per side [m]	L_s	0.48
Wing chord length [m]	L_c	0.22
Propeller wheelbase [m]	L_{pp}	0.32

components bound to the T-shaped carbon fiber tubes, including all driving servos, as these components are fixed. Correspondingly, the mass of a single wing and that of a single vectoring rotor exclude their respective driving servos, representing only the mass of the movable components. This definition is formulated to facilitate subsequent multibody dynamics modeling. Hereinafter, the main airframe structure is referred to as the airframe for brevity.

2.2. Actuation degree-of-freedom allocation

[Fig. 2](#) introduces the definitions of each coordinate system of the PW-TUAV and the actuation degree-of-freedom allocation method. Among these, [Fig. 2\(a\)](#) defines the body coordinate system in hover mode or transition phase, whose origin O_b coincides with the aircraft's center of gravity. The X_b -axis points to the front of the fuselage, the Y_b -axis points to the right side of the fuselage, and the Z_b -axis points to the underside of the fuselage. The symbols of the three-axis Euler angles are defined as ϕ , θ , and ψ respectively. [Fig. 2\(b\)](#) illustrates the definition method of the wing coordinate system, taking the right wing as an example. Its origin O_{w2} is the intersection of the wing rotation axis and the leading-edge axis. The Y_{w2} -axis coincides with the leading-edge axis, the Z_{w2} -axis coincides with the wing rotation axis, and the X_{w2} -axis is determined by the right-hand rule. The included angle between the Y_{w2} -axis and the Y_b -axis is defined as the wing phase angle λ_2 , whose positive direction is the counterclockwise direction around the Z_{w2} -axis. [Fig. 2\(d\)](#) presents the definition method of the vectoring rotor coordinate system, taking the left vectoring rotor mechanism as an example. Its origin O_{p1} is the intersection of the propeller rotation axis and the vectoring mechanism rotation axis. The Y_{p1} -axis coincides with the vectoring mechanism rotation axis, the Z_{p1} -axis coincides with the propeller rotation axis, and the X_{p1} -axis is determined by the right-hand rule. The included angle between the Z_{p1} -axis and the Z_b -axis is defined as the rotor tilt angle ξ_1 ,

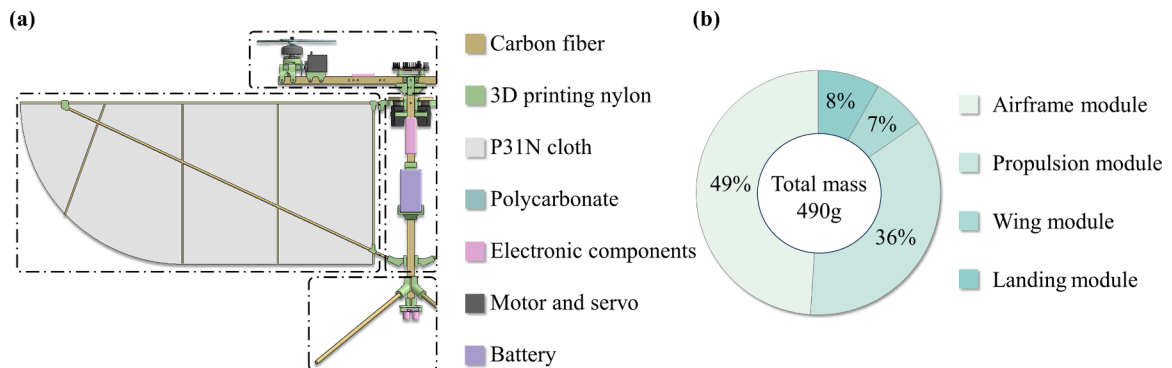


Fig. 1. Layout introduction of the PW-TUAV prototype. (a) Front view of the 3D model to showcase the structural composition and material properties. (b) Mass distribution diagram.

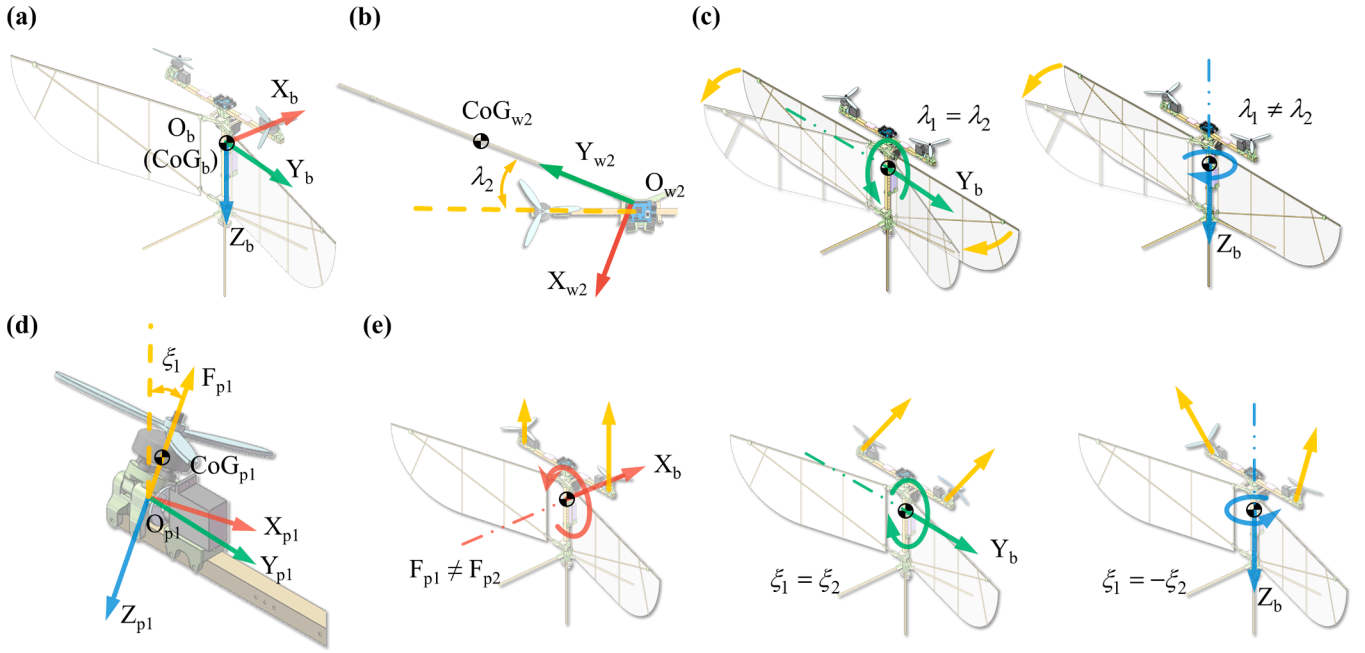


Fig. 2. Coordinate system definition and introduction of actuation degrees of freedom. (a) Definition of the body coordinate system. (b) Definition of the wing coordinate system, taking the right wing as an example. (c) Schematic diagram of the auxiliary control torque principle induced by wing phase shifting. (d) Definition of the vectoring rotor coordinate system, taking the left one as an example. (e) Schematic diagram of the control torque generation principle for the vectoring rotor.

whose positive direction is the counterclockwise direction around the Y_{p1} -axis.

Fig. 2(c) describes the actuation principle of the phase-shifting wing. When the phase angles of the wings on both sides change symmetrically, a pitching moment in the body coordinate system is mainly generated; when the phase angles of the wings on both sides change asymmetrically, a yawing moment in the body coordinate system is mainly generated. The actual mechanical mechanism is highly complex, involving multiple effects such as center of gravity variation, aerodynamic damping, and projected area variation, which includes both active actuation torque and passive aerodynamic torque variation. However, it is noteworthy that the torque directions induced by these effects are consistent, so the schematic diagram in Fig. 2(c) can be used for qualitative description. A detailed analysis will be presented in Section 3.3.4. The wing can perform phase angle variation with a maximum travel range of approximately $0\text{--}80^\circ$, driven by a 25 g digital servo model PTK 8515 MG-D. The driving voltage of the servo is 7.4 V, with a corresponding stall torque of approximately 1.35 N·m and an operating speed of approximately 0.1 s per 60° .

Fig. 2(e) is the schematic diagram illustrating the torque generation principle of the vectoring rotor mechanism. The PW-TUAV prototype is equipped with two 5-inch propellers of model F5146 manufactured by T-motor. Powered by a 3S RC lithium battery, a single propeller can provide a maximum thrust of approximately 400 g. The direction of the propeller thrust vector can be adjusted within a maximum range of $\pm 50^\circ$ by a 9 g digital servo designed by Henge Company. The driving voltage of these two servos is 5 V, with a corresponding stall torque of approximately 0.16 N·m and an operating speed of approximately 0.08 s per 60° . Differential control of the rotors on both sides enables the generation of a roll moment in hover mode or a yaw moment in cruise mode within the body coordinate system. When the rotors on both sides tilt to the same side simultaneously, a pitching moment is generated in all flight modes. When the rotors on both sides tilt in opposite directions, a yaw moment is generated in hover mode or a roll moment in cruise mode. It is worth noting that since the control method adopted in this study is designed based on the continuous SO(3) space, both the feedback attitude matrix and the desired attitude matrix are unique at the same moment. The only difference is that the rotation sequence for

generating the desired attitude matrix is switched according to the hover or cruise mode, and this switching is solely for the convenience of manual remote control. The switching is executed during the transition phase and is completely seamless; therefore, the definition of the body coordinate system remains unchanged regardless of whether the aircraft is in hover or cruise mode during dynamic analysis.

2.3. Mechanism design and analysis of phase-shifting wings

Active wing motion control technology inspired by bionics has been widely adopted in the field of flapping-wing UAVs [44,45]. However, since the propulsive force of flapping-wing UAVs is entirely generated by the combination of wing heaving and pitching motions, most of these UAVs perform fixed reciprocating motions at relatively high flapping frequencies [46,47]. Only a small number of flapping-wing prototypes directly driven by digital servos involve research on aircraft attitude control by altering wing motion patterns [41–43]. A common characteristic of such flapping-wing prototypes is their ultra-lightweight, typically weighing around 100 g. The fundamental reason is that micro servos can hardly achieve both high rotational speed and high torque simultaneously. In this study, the PW-TUAV draws on the active wing motion control technology from the flapping-wing UAV field. The key difference, however, lies in the fact that the PW-TUAV is equipped with a propeller propulsion module. This means that it no longer needs to maintain continuous high-speed flapping-wing motion; instead, it can modify the aircraft's aerodynamic characteristics to assist attitude control merely by adjusting the wing phase angles, resulting in a more explicit and observable control mechanism. At the same time, though, the heavier fuselage weight and higher flight speed pose new challenges to the driving torque performance of micro servos.

As a rough estimate, even under the highly ideal condition of steady and constant-speed cruise mode, the aerodynamic force perpendicular to the wing surface borne by a single wing of the aircraft is at least half of the total fuselage weight, with an aerodynamic moment arm nearly equal to half the wingspan of the single wing. The magnitude of this moment increases quadratically with the scaling-up of the prototype and is persistently present. A direct drive scheme using servos is obviously energy-intensive and inefficient. Therefore, the PW-TUAV adopts a

Grashoff mechanism (also referred to as the four-bar linkage mechanism for short) to address this problem more appropriately. The specific implementation and analysis process are illustrated in Fig. 3.

Fig. 3(a) illustrates the actuation principle of the wing drive mechanism in its partial enlarged view. Among these points, D and A are fixed to the airframe, while point B can perform a relative rotation with a maximum stroke of approximately 160° around point A. The motion of point B drives the motion of point C and transmits this motion to the wing fixture component CDE, thereby driving the wing to rotate. L_1 is defined as the frame length, L_2 as the crank length, L_3 as the connecting rod length, and L_4 as the rocker length. L_5 and L_6 are fixed integrally with L_4 . The parameter design is mainly based on the convenience of assembly; thus, this is a rough but feasible design with potential for further optimization. In this example, the link length parameters are given in Table 2. The included angle between each link L_i and the x-axis of the local planar coordinate system is defined as θ_i . In this case, θ_1 is a known constant and θ_2 is a known variable. θ_3 and θ_4 vary with θ_2 and need to be solved according to the vector loop equation. θ_5 is the actual angle of the wing, namely the phase angle λ , and the difference between θ_5 and θ_4 is a constant value.

The detailed derivation of the vector loop equations is presented in Appendix C, which includes the transfer relationships of the angles, angular velocities, and angular accelerations between the linkages.

When the input θ_2 ranges from -115° to 45° , the position variations of each link are illustrated in Fig. 3(b) to Fig. 3(d). As can be seen from Fig. 3(b), the initial position of the mechanism is designed to be very close to the back-drive dead zone, where the wing dihedral angle is nearly 0° . This means that when the wing faces the wind with the maximum projected area, the undesirable torque generated along the Z_w -axis is completely counteracted by the structural force of the drive mechanism. In Fig. 3(e), the angle transfer relationship of the four-bar linkage mechanism is presented more intuitively with θ_2 as the abscissa.

The angular velocity and angular acceleration transfer relationships

Table 2
Linkage length parameters.

Parameters	Symbol	Value
Frame length [mm]	L_1	24.6
Crank length [mm]	L_2	15
Connecting rod length [mm]	L_3	32
Rocker length [mm]	L_4	21
Structure length [mm]	L_5	20
Structure length [mm]	L_6	6.75

of the four-bar linkage mechanism are illustrated in Fig. 3(f) and Fig. 3(g), respectively. Since points C, D, and E are part of the same component structure, links L_4 and L_5 have equal angular velocities and angular accelerations.

Next, the torque transfer relationship between the components of the four-bar linkage mechanism is analyzed. Given that the purpose is to verify whether the lock torque of the servo is sufficient, the problem can be simplified to a statics analysis. Neglecting the mass of link L_3 , the forces at points B and C are equal in magnitude, opposite in direction, and collinear with the link; let the magnitude of this force be F_{L3} . Considering a known aerodynamic torque M_a , the torque equilibrium equation for component CDE is as follows:

$$M_a - F_{L3}L_4\sin(\theta_3 - \theta_4) = 0 \quad (1)$$

The torque equilibrium of link L_2 is next examined, where the torque exerted by F_{L3} is balanced by the servo drive torque M_s :

$$M_s - F_{L3}L_2\sin(\theta_3 - \theta_2) = 0 \quad (2)$$

The basic torque transfer coefficient is defined as the ratio of the output torque of the servo to the external load torque acting on the wing, that is:

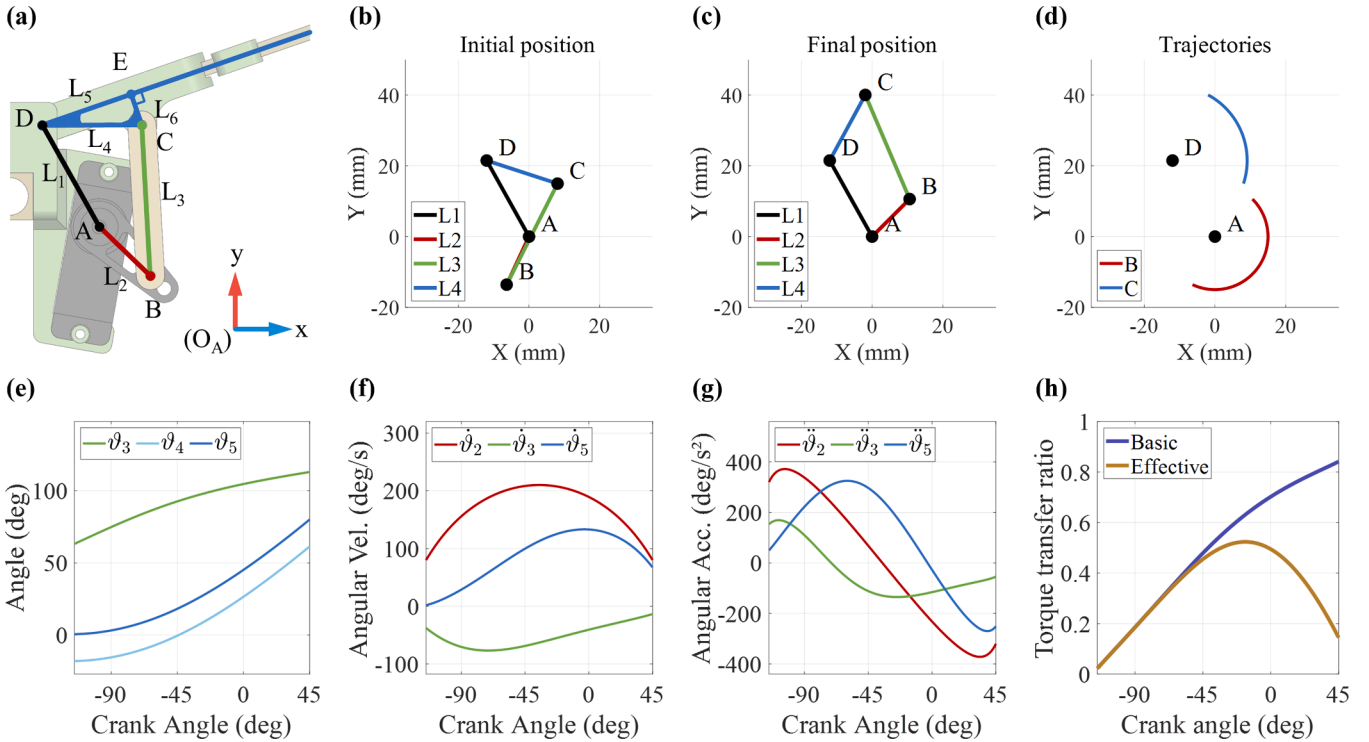


Fig. 3. Kinematic analysis of the phase-shifting wing. (a) Motion schematic: A deformable wing phase angle adjustment mechanism designed based on the four-bar linkage mechanism. (b) Initial positions of components: The position where points A, B and C are nearly collinear, with the wing dihedral angle (i.e., the included angle between DE and the x-axis) being 0° . (c) Limit motion position: Determined by the maximum motion stroke angle of the micro servo. (d) Motion trajectory from the initial position to the limit position. (e) Angle transfer relationship. (f) Angular velocity transfer relationship. (g) Angular acceleration transfer relationship. (h) Aerodynamic torque transfer relationship.

$$\frac{M_s}{M_a} = \frac{L_2 \sin(\theta_3 - \theta_2)}{L_4 \sin(\theta_3 - \theta_4)} \quad (3)$$

In addition, considering the actual cruise mode, when the wing angle is elevated, the effective projected area of the wing in the forward direction decreases accordingly. According to the dynamic pressure formula, the relationship between the actual aerodynamic torque M_a and the aerodynamic torque M_{a0} when the wing is at a horizontal angle of 0° is as follows:

$$\frac{M_a}{M_{a0}} = \cos\theta_5 \quad (4)$$

Therefore, the effective torque transfer coefficient can be expressed as:

$$\frac{M_s}{M_{a0}} = \frac{L_2 \sin(\theta_3 - \theta_2)}{L_4 \sin(\theta_3 - \theta_4)} \cos\theta_5 \quad (5)$$

Fig. 3(h) shows the variation trends of the basic torque transfer coefficient and effective torque transfer coefficient with crank angle. It can be observed that the torque transfer coefficient tends to zero near the initial position illustrated in Fig. 3(b). This indicates that when the aircraft cruises with the wing at maximum projected area, the aerodynamic torque about the Z_w -axis is almost completely counteracted by the structural force of the drive mechanism, thereby avoiding the problems of insufficient holding torque and energy waste that would occur if the wing were directly driven by the servo. The maximum value of the effective torque transfer coefficient across all operating conditions is also <0.5 , which means the driving performance of the servo is doubled. In the design case of the PW-TUAV prototype, the aerodynamic torque about the Z_w -axis acting on a single wing during cruise mode can be roughly estimated as half the weight multiplied by half the span of the single wing. Based on the parameters in Table 1, the calculated result is 1.15 N·m. The PTK 8515 MG-D digital servo motor used to drive the wing has a stall torque of approximately 1.35 N·m at a voltage of 7.4 V. If a direct servo drive scheme were adopted, the margin of the holding torque would be extremely narrow. The servo would easily lose its locking capability under gust disturbances, leading to uncontrollable wing phase and subsequent instability of the flight control system. In contrast, the proposed four-bar linkage transmission scheme increases the effective holding torque to >2.7 N·m. This is a substantial value for a micro digital servo weighing around 20 g, providing a sufficient margin to maintain the stable and controllable wing angle.

3. Dynamic modeling methodology

Based on the proposed PW-TUAV configuration described in the preceding section, this section will establish the aircraft's dynamic model and analyze the special response characteristics of the drive mechanism. First, the aerodynamic characteristics of the phase-shifting wing are analyzed; then, a multi-body dynamic model accounting for the position variations of the wings and rotors is established; finally, the dynamic mechanisms of the vectoring rotor mechanism and the phase-shifting wing mechanism are analyzed separately.

3.1. Aerodynamic characteristics of phase-shifting wings

In this subsection, we conduct a detailed analysis of the wing aerodynamic mechanism for the PW-TUAV based on its motion characteristics. Compared with conventional aircraft dynamics modeling approaches, the modeling theory proposed here specifically addresses two key issues unique to the PW-TUAV's motion characteristics: (a) The wings are perpendicular to the ground in VTOL mode, while nearly parallel to the ground in cruise mode. This results in the wing aerodynamic center not remaining fixed over such a wide range of angle of attack variations. (b) The wings have the degree of freedom to actively adjust their phase angle to coordinate with the vectoring rotor

mechanism, forming a control mechanism of the SC. This rotational motion generates unsteady aerodynamic forces and also affects the position of the aerodynamic center in real time.

According to thin airfoil theory, for a planar airfoil, the chordwise position of the aerodynamic center lies at 25% of the chord length from the leading-edge axis when the angle of attack is below 15° . When the angle of attack exceeds 45° , the chordwise position of the aerodynamic center shifts to 50% of the chord length from the leading-edge axis. When the angle of attack ranges between 15° and 45° , the aerodynamic center becomes unstable and is situated between the two limit positions. To ensure the static stability margin during cruise mode, the aircraft's center of gravity needs to be placed forward, generally at around 20% of the chord length from the leading-edge axis. This configuration is very similar to that of flying-wing UAVs without additional tail surfaces [48]. However, tail-sitter UAVs have an extra VTOL capability compared with flying-wing UAVs. During the VTOL phase, the aerodynamic center remains at approximately 50% of the chord length from the leading-edge axis throughout almost the entire process, while the center of gravity position does not change during flight. This leads to a significant aerodynamic pitching moment, which is the primary reason for the insufficient stability of tail-sitter UAVs during the VTOL phase. Therefore, the variation of the aerodynamic center position with flight states must be taken into account when establishing the aerodynamic model.

However, such aerodynamic characteristics are a typical unsteady flow phenomenon at low Reynolds numbers. Establishing a systematic fluid-structure interaction model is highly complex and requires extremely substantial computational resources. Given the strong similarity between this mode of motion and flapping-wing motion, reference can be made to certain aerodynamic modeling methods for flapping-wing UAVs. The widely adopted reduced-order aerodynamic model [51,52] based on the quasi-steady (QS) assumption [49,50] is also applicable to the PW-TUAV, though appropriate adjustments tailored to the characteristics of the PW-TUAV are required in practical applications. Special aerodynamic mechanisms with negligible effects can be appropriately simplified or even neglected. Meanwhile, the calculation of aerodynamic torque needs to be refined, since the characteristic velocity of the wing is distributed unevenly along the chordwise direction.

First, the aspects to be simplified are clarified. Since the wings of the PW-TUAV do not need to generate thrust, there is no requirement to rely on flexural deformation to achieve the combination of heaving and pitching motions of the wings, which makes its aerodynamic mechanism far less complex than that of actual flapping-wing UAVs. Special aerodynamic mechanisms such as rotational circulation, added mass, wing-wake interaction, and clap and fling can all be neglected [53]; only the translational circulatory force under the condition of time-varying local characteristic velocity needs to be considered.

Next, the aspects requiring refinement are elaborated. Most studies on flapping-wing UAVs still focus on how to characterize the aerodynamic forces of wings under constant-speed cruise conditions; thus, the simplified blade element model combined with the QS model is widely adopted. In contrast, the flight mode of the PW-TUAV is more complex. Particularly during the transition phase, it needs to perform maneuvering actions with a wide range of angle-of-attack variations, which results in the non-uniform chordwise distribution of the characteristic velocity of the wing. The blade element model, which only discretizes the wing uniformly along the spanwise direction, cannot adequately describe this mechanism. It is therefore necessary to discretize the wing again uniformly along the chordwise direction to develop a finite element model that can capture more detailed characteristics. Of course, the finite element model still has certain limitations. Existing research [54] has demonstrated that this empirical formula-based algorithm can accurately predict the aerodynamic forces of the aircraft, but it may not be applicable to aerodynamic torque analysis. This is because the actual chordwise distribution of vortex strength is not entirely determined by the characteristic velocity distribution. The solution adopted in this study is to decompose the

aerodynamic torque generated by the wing's translational and rotational motions into separate components, which are then superposed to obtain the total torque. The aerodynamic torque induced by translational motion is easy to characterize, since the aerodynamic center can be directly determined using thin airfoil theory. The analysis of rotational motion is slightly more complex. First, the aerodynamic force generated by rotational motion cannot be directly calculated using the dynamic pressure formula. This is because the dynamic pressure formula is based on the resultant characteristic velocity, and changes in the direction of the characteristic velocity will also lead to variations in the angle of attack. Instead, this aerodynamic force must be derived by subtracting the aerodynamic force from pure translational motion from the resultant aerodynamic force, thereby accurately establishing the correct response relationship under different incoming flow velocity conditions. Second, the aerodynamic force generated by rotational motion does not have a stable aerodynamic center. Therefore, the corresponding aerodynamic torque needs to be obtained via double integration using the finite element model.

The detailed modeling derivation is presented below, divided into aerodynamic force analysis and aerodynamic moment analysis, which are elaborated in SubSections 3.1.1 and 3.1.2, respectively.

3.1.1. Aerodynamic force analysis

Based on the finite element model, the characteristic velocity U_e^{wi} of the wing element mesh defined in the wing frame is expressed as [55]

$$U_e^{wi} = U_{tran,e}^{wi} + U_{rot,e}^{wi} \quad (6)$$

$$U_{tran,e}^{wi} = R_{b,wi} \left(\mathbf{v}_b^b + \mathbf{v}_{gust}^b \right) \quad (7)$$

$$U_{rot,e}^{wi} = R_{b,wi} \left(\omega_b^b \times \mathbf{r}_{ob,Coe}^b \right) + \omega_{wi,b}^{wi} \times \mathbf{r}_{owi,Coe}^{wi} \quad (8)$$

where $U_{tran,e}^{wi}$ denotes the pure translational velocity of the wing element mesh, and $U_{rot,e}^{wi}$ denotes the velocity of the wing element mesh induced by the rotation of the airframe or the wing itself. $R_{b,wi}$ is the rotation matrix transforming from the airframe frame to the i th wing frame; its expression can be derived as shown in Eq. (9) based on Fig. 2, with its inverse matrix denoted as $R_{wi,b}$. $\mathbf{v}_b^b$ is the linear velocity of the airframe frame, ω_b^b is the angular velocity of the airframe frame, and $\mathbf{v}_{gust}^b$ is the external disturbance gust velocity. $\omega_{wi,b}^{wi}$ is the angular velocity of the i th wing frame relative to the airframe frame, which can be expressed as shown in Eq. (10). $\mathbf{r}_{ob,Coe}^b$ is the coordinate of the geometric center of the wing element in the airframe frame, which can be derived via Eq. (11). $\mathbf{r}_{owi,Coe}^{wi}$ is the coordinate of the geometric center of the wing element in the i th wing frame, varying with the index relationship. $\mathbf{r}_{ob,owi}^b$ is the coordinate of the origin of the i th wing frame in the airframe frame, which can be expressed as shown in Eq. (12). In the design case of the PW-TUAV prototype, L_{xw} is approximately 0.014 m; L_{yw} is negligible and thus can be treated as zero; L_{zw} can be approximated as one-fifth of the wing chord length L_c , i.e., 0.044 m.

$$\begin{cases} R_{wi,b} = R_z[i\pi + (-1)^i \lambda_i] \\ R_{b,wi} = R_z[i\pi + (-1)^{i+1} \lambda_i] \end{cases} \quad (9)$$

$$\omega_{wi,b}^{wi} = [\dot{R}_{wi,b} (R_{wi,b})^T]^\vee \quad (10)$$

$$\mathbf{r}_{ob,Coe}^b = \mathbf{r}_{ob,owi}^b + R_{wi,b} \mathbf{r}_{owi,Coe}^{wi} \quad (11)$$

$$\mathbf{r}_{ob,owi}^b = [-L_{xw}, (-1)^i L_{yw}, -L_{zw}]^T \quad (12)$$

In Eq. (10), $R_z(\cdot)$ denotes the rotation matrix about the z -axis, and the $(\cdot)^\vee$ operator represents the operation that maps a skew-symmetric matrix back to its vector form [56]. Owing to the special form of $R_{b,wi}$,

$\omega_{wi,b}^{wi}$ can be simplified to:

$$\omega_{wi,b}^{wi} = [0, 0, (-1)^i \dot{\lambda}_i]^T \quad (13)$$

The local angle of attack α_e at the wing element mesh is defined as the angle between the projection of the characteristic velocity U_e^{wi} onto the $X_{wi}O_{wi}Z_{wi}$ plane and the negative direction of the Z_{wi} -axis:

$$\alpha_e = \text{atan2}(-U_e^{wi}(1), -U_e^{wi}(3)) \quad (14)$$

where $U_e^{wi}(1)$ denotes the component of U_e^{wi} along the X_{wi} -axis, and $U_e^{wi}(3)$ denotes the component of U_e^{wi} along the Z_{wi} -axis.

The calculation method for the aerodynamic circulation component is referenced from the widely used Dickinson model [57]. Since the analysis object is a planar thin airfoil, the effect of airflow viscous force can be neglected, and $U_e^{wi}(2)=0$ is set to simplify the calculation of aerodynamic forces in the longitudinal channel. The normal aerodynamic force at any wing element mesh can be expressed as:

$$f_{N,e} = \frac{1}{2} \rho_{air} C_{N,e} (U_e^{wi})^2 dS_e \quad (15)$$

where ρ_{air} denotes the air density, S_e denotes the effective projected area of the wing element, and $C_{N,e}$ is the translational circulation coefficient, which is a function of the local angle of attack α_e . It can be referenced to the general model proposed by Nabawy et al. [58]:

$$\begin{cases} C_{L,e} = \left(\frac{0.5 C_{La,2d}}{E + \frac{k C_{La,2d}}{\pi AR}} \right) \sin 2\alpha_e \\ C_{N,e} = \left(\frac{C_{La,2d}}{E + \frac{k C_{La,2d}}{\pi AR}} \right) \sin \alpha_e \end{cases} \quad (16)$$

where $C_{La,2d}$ can take a value of 5.16 rad^{-1} according to the study by Okamoto et al. [59]. The parameter E is the edge correction factor from the lifting-line theory proposed by Jones et al. [60], defined as the ratio of the wing semiperimeter to its span. AR denotes the aspect ratio of a single wing, and k is a coefficient used to correct for the efficiency discrepancy between the assumed ideal uniform downwash distribution and the actual downwash distribution. Based on the study by Nabawy et al. [61], k is set to 1.5 given that the research object is bird-scale.

Integrating $f_{N,e}$ over the area domain yields the aerodynamic force expressed in the wing frame:

$$\mathbf{F}_{a,wi}^{wi} = \left[\int_A f_{N,e}, 0, 0 \right]^T \quad (17)$$

3.1.2. Aerodynamic moment analysis

By analogy, for any wing element mesh, its angle of attack under pure translational velocity can be expressed as:

$$\alpha_0 = \text{atan2}(-U_{tran,e}^{wi}(1), -U_{tran,e}^{wi}(3)) \quad (18)$$

Since the pure translational velocity is consistent across all wing element meshes, this common characteristic angle of attack is denoted for short as α_0 . Substituting α_0 into Eq. (16), the corresponding aerodynamic coefficient C_{0N} is calculated. Subsequently, based on the dynamic pressure formula, the aerodynamic force induced by pure translational velocity for any wing element mesh can be expressed as:

$$f_{tran,e} = \frac{1}{2} \rho_{air} C_{0N} (U_{tran,e}^{wi})^2 dS_e \quad (19)$$

The aerodynamic force on the wing element mesh induced by rotational motion under different incoming flow velocity conditions can be

obtained by taking the difference:

$$f_{rot,e} = f_{N,e} - f_{tran,e} \quad (20)$$

In the wing frame, the aerodynamic torque relative to the center of mass induced by rotational motion for the wing element mesh can be expressed as:

$$\tau_{rot,e}^{wi} = (r_{owi,CoA}^{wi} - r_{owi,mwi}^{wi}) \times [f_{rot,e}, 0, 0]^T \quad (21)$$

where $r_{owi,mwi}^{wi}$ is the position of its own center of mass in the wing frame, which can be simplified in the model as:

$$r_{owi,mwi}^{wi} = \left[0, \frac{L_s}{2}, \frac{L_c}{2} \right]^T \quad (22)$$

Integrating $\tau_{rot,e}^{wi}$ over the area domain yields the total aerodynamic torque of the entire wing induced by rotational motion:

$$\tau_{rot,wi}^{wi} = \int_A \tau_{rot,e}^{wi} \quad (23)$$

Meanwhile, the aerodynamic force of the wing induced by pure translational motion, expressed in the wing frame, can be written as:

$$F_{tran,wi}^{wi} = \left[\int_A f_{tran,e}, 0, 0 \right]^T \quad (24)$$

The corresponding aerodynamic torque can be expressed as:

$$\tau_{tran,wi}^{wi} = (r_{owi,CoA}^{wi} - r_{owi,mwi}^{wi}) \times F_{tran,wi}^{wi} \quad (25)$$

where $r_{owi,CoA}^{wi}$ is the position of the aerodynamic center in the wing frame, determined according to thin airfoil theory. A linearized model simplification approach is adopted for the turbulent flow region, which can be expressed as:

$$r_{owi,CoA}^{wi} = \left[0, \frac{L_s}{2}, L_{zCoA} \right]^T \quad (26)$$

$$L_{zCoA} = \begin{cases} \frac{L_c}{4}, & |\alpha_0| < \frac{\pi}{12} \\ \frac{L_c}{8} + \frac{3L_c}{2\pi} |\alpha_0|, & \frac{\pi}{12} < |\alpha_0| < \frac{\pi}{4} \\ \frac{L_c}{2}, & \frac{\pi}{4} < |\alpha_0| < \frac{\pi}{2} \end{cases} \quad (27)$$

Finally, the total aerodynamic torque of the entire wing section, obtained via superposition, can be expressed as:

$$\tau_{a,wi}^{wi} = \tau_{rot,wi}^{wi} + \tau_{tran,wi}^{wi} \quad (28)$$

3.2. Multibody dynamics modeling

In this subsection, a multibody dynamics modeling approach is adopted to integrate the dynamics of the end components into the aircraft system, so as to accurately characterize its overall dynamic response.

Based on the relevant parameters in Table 2, the PW-TUAV is equipped with a pair of vectoring rotors and a pair of phase-shifting wings, with a total mass of approximately 108 g. The mass of these relative-motion-generating mechanisms accounts for about 22% of the total mass of the entire aircraft, and their centers of mass are all far offset from the aircraft's center of mass, resulting in a long moment arm for inertial forces. Therefore, the use of a single rigid-body model may lead to significant modeling errors; for instance, the non-minimum phase characteristics of the vectoring rotor mechanism [62,63] may be

inaccurately characterized by a single rigid-body model. It is thus imperative to establish a multi-body dynamic model. In the PW-TUAV, both the vectoring rotors and phase-shifting wings are first-stage end components, and there is no multi-stage transmission in the entire system. For this reason, the recursive Newton–Euler method offers a more convenient and intuitive approach.

The multibody dynamics model is established in six steps below. The first four steps are outward kinematic recursions, and the latter two are inward dynamic recursions. The detailed derivation is as follows.

Step1: Angular velocity recursion

The angular velocity of the end-effector mechanism is equal to the superposition of the airframe angular velocity and the relative motion angular velocity, with the calculation expressed in the airframe frame as:

$$\omega_{pi}^b = \omega_b^b + \omega_{pi,b}^b \quad (29)$$

$$\omega_{wi}^b = \omega_b^b + \omega_{wi,b}^b \quad (30)$$

where ω_{pi}^b denotes the resultant angular velocity of the i th vectoring rotor mechanism, ω_{wi}^b denotes the resultant angular velocity of the i th phase-shifting wing, ω_b^b denotes the airframe angular velocity, $\omega_{pi,b}^b$ denotes the rotational velocity of the i th vectoring rotor frame relative to the airframe frame, and $\omega_{wi,b}^b$ denotes the rotational velocity of the i th wing frame relative to the airframe frame. $\omega_{wi,b}^b$ has been derived in Eq. (13). Since its direction is about the Z_{wi} -axis and the Z_{wi} -axis is always collinear with the Z_b -axis, we have:

$$\omega_{wi,b}^b = \omega_{wi,b}^{wi} = [0, 0, (-1)^i \dot{\lambda}_i]^T \quad (31)$$

By analogy, the derivation of $\omega_{pi,b}^b$ is almost identical to that of $\omega_{wi,b}^b$. The only difference is that $\omega_{pi,b}^b$ is the angular velocity about the Y_{pi} -axis, and the positive direction of the rotor tilt angle ξ_i is defined consistently across both sides—unlike the phase angle λ_i of the wing, whose positive direction is defined oppositely for the left and right sides. Therefore, there is no need to specify distinct positive directions for the two sides. The detailed derivation process is as follows:

$$\begin{cases} R_{pi,b} = R_y(\xi_i) \\ R_{b,pi} = R_y(-\xi_i) \end{cases} \quad (32)$$

$$\omega_{pi}^{pi} = [\dot{R}_{b,pi}(R_{b,pi})^T]^\vee \quad (33)$$

$$\omega_{pi,b}^b = \omega_{pi}^{pi} = [0, \dot{\xi}_i, 0]^T \quad (34)$$

Step2: Angular acceleration recursion

Differentiating the angular velocity equations with respect to time yields the angular acceleration equations:

$$\dot{\omega}_{pi}^b = \dot{\omega}_b^b + \dot{\omega}_{pi,b}^b + \omega_b^b \times \omega_{pi,b}^b \quad (35)$$

$$\dot{\omega}_{wi}^b = \dot{\omega}_b^b + \dot{\omega}_{wi,b}^b + \omega_b^b \times \omega_{wi,b}^b \quad (36)$$

where $\dot{\omega}_{pi}^b$ is the resultant angular acceleration of the i th vectoring rotor mechanism, $\dot{\omega}_b^b$ is the angular acceleration of the airframe, $\dot{\omega}_{pi,b}^b$ is the relative angular acceleration, and $\omega_b^b \times \omega_{pi,b}^b$ is the Coriolis correction term. The definitions of physical quantities for the phase-shifting wing mechanism are consistent with those for the vectoring rotor mechanism.

Step3: Linear velocity recursion

The linear velocity of the end-effector mechanism is defined as the motion velocity of its center of mass. The linear velocity equation of the end-effector in the airframe frame is given by:

$$v_{pi}^b = v_b^b + \omega_b^b \times r_{ob,opi}^b + \omega_{pi,b}^b \times r_{opi,mpi}^b \quad (37)$$

$$\mathbf{v}_{wi}^b = \mathbf{v}_b^b + \omega_b^b \times \mathbf{r}_{ob,owi}^b + \omega_{wi,b}^b \times \mathbf{r}_{owi,mwi}^b \quad (38)$$

The physical meanings of each term in the equation are as follows: the first term represents the transport translational velocity of the end-effector mechanism's center of mass induced by the airframe's translational velocity; the second term denotes the transport tangential velocity of the end-effector mechanism's center of mass caused by the airframe's rotation; the third term is the linear velocity generated by the rotation of the end-effector mechanism's center of mass around the drive shaft. The definitions of each symbol are given below: $\mathbf{r}_{ob,owi}^b$ is the position coordinate of the origin of the i th wing frame in the airframe frame, which has been defined in Eq. (12). $\mathbf{r}_{ob,opi}^b$ is the position coordinate of the origin of the i th vectoring rotor frame in the airframe frame, whose definition is similar to that of $\mathbf{r}_{ob,owi}^b$. It is also a fixed value determined by design parameters and can be expressed as shown in Eq. (39). In the design case of the PW-TUAV prototype, L_{xp} is negligible and thus can be treated as zero; L_{yp} is equal to half the propeller wheelbase, i. e., 0.16 m; and L_{zp} is approximately 0.09 m.

$\mathbf{r}_{opi,mpi}^b$ and $\mathbf{r}_{owi,mwi}^b$ are the expressions of the end-effector mechanism's center of mass position in its local frame transformed into the airframe frame. This transformation is performed to ensure that all calculations are conducted in the same coordinate frame, and they can be obtained via the coordinate transformation relationships shown in Eq. (40) and Eq. (41). Among them, $\mathbf{r}_{owi,mwi}^w$ has been defined in Eq. (22), while $\mathbf{r}_{opi,mpi}^p$ can be expressed in the form of Eq. (42). Here, L_{mp} is the distance from the center of mass of the vectoring rotor mechanism to its rotating shaft, which is approximately 0.022 m in the design case of the PW-TUAV prototype.

$$\mathbf{r}_{ob,opi}^b = [L_{xp}, (-1)^i L_{yp}, -L_{zp}]^T \quad (39)$$

$$\mathbf{r}_{opi,mpi}^b = \mathbf{R}_{pi,b} \mathbf{r}_{opi,mpi}^p \quad (40)$$

$$\mathbf{r}_{owi,mwi}^b = \mathbf{R}_{wi,b} \mathbf{r}_{owi,mwi}^w \quad (41)$$

$$\mathbf{r}_{opi,mpi}^p = [0, 0, -L_{mp}]^T \quad (42)$$

Step4: Linear acceleration recursion

Differentiating the linear velocity equation with respect to time using the chain rule yields the linear acceleration equation. Given the large number of terms in the original equation, the result can be grouped by physical meaning into:

$$\dot{\mathbf{v}}_{pi}^b = \mathbf{a}_{cn,pi}^b + \mathbf{a}_{rel,pi}^b + \mathbf{a}_{cor,pi}^b \quad (43)$$

where $\mathbf{a}_{cn,pi}^b$ denotes the transport acceleration, $\mathbf{a}_{rel,pi}^b$ denotes the relative acceleration, and $\mathbf{a}_{cor,pi}^b$ denotes the Coriolis acceleration. Their complete expressions are given by:

$$\begin{cases} \mathbf{a}_{cn,pi}^b = \dot{\mathbf{v}}_b^b + \omega_b^b \times \mathbf{v}_b^b + \dot{\omega}_b^b \times \mathbf{r}_{ob,opi}^b + \omega_b^b \times (\omega_b^b \times \mathbf{r}_{ob,opi}^b) \\ \mathbf{a}_{rel,pi}^b = \dot{\omega}_{pi,b}^b \times \mathbf{r}_{opi,mpi}^b + \omega_{pi,b}^b \times (\omega_{pi,b}^b \times \mathbf{r}_{opi,mpi}^b) \\ \mathbf{a}_{cor,pi}^b = \omega_{pi,b}^b \times (\omega_b^b \times \mathbf{r}_{opi,mpi}^b) + \omega_b^b \times (\omega_{pi,b}^b \times \mathbf{r}_{opi,mpi}^b) \end{cases} \quad (44)$$

For the phase-shifting wing mechanism, the corresponding equations are identical to those of the vectoring rotor mechanism, with the only modification being the replacement of the subscript pi with wi .

Step5: End-effector Newton-Euler equations

In this step, it is necessary to calculate the interaction forces and moments required to generate the accelerations and angular accelerations derived in the kinematic analysis, starting from the end components. Since the expression form of forces is relatively simple, Newton's equations are established directly in the airframe frame. In this way, the obtained internal constraint forces can be directly applied to the

airframe force analysis in Step 6 without further coordinate transformation. However, Euler's equations need to be established in the end reference frame, because this avoids the complex solution of the time-varying inertia tensor.

Newton's equations for the end components can be expressed as:

$$m_p (\dot{\mathbf{v}}_{pi}^b + \omega_{pi}^b \times \mathbf{v}_{pi}^b) = \mathbf{F}_{r,pi}^b + \mathbf{F}_{p,pi}^b + m_p \mathbf{g}^b \quad (45)$$

$$m_w (\dot{\mathbf{v}}_{wi}^b + \omega_{wi}^b \times \mathbf{v}_{wi}^b) = \mathbf{F}_{r,wi}^b + \mathbf{F}_{a,wi}^b + m_w \mathbf{g}^b \quad (46)$$

where $\mathbf{F}_{r,pi}^b$ and $\mathbf{F}_{r,wi}^b$ denote the constraint forces exerted by the airframe on the end components, and they are the unknowns to be solved in this step. $\mathbf{F}_{p,pi}^b$ is the thrust vector of the propeller expressed in the airframe frame, $\mathbf{F}_{a,wi}^b$ is the vector form of the wing aerodynamic force in the airframe frame, and $\mathbf{g}^b$ is the gravitational acceleration vector in the airframe frame. These quantities can be obtained via coordinate transformation respectively:

$$\mathbf{F}_{p,pi}^b = \mathbf{R}_{pi,b} [0, 0, -F_i]^T \quad (47)$$

$$\mathbf{F}_{a,wi}^b = \mathbf{R}_{wi,b} \mathbf{F}_{a,wi}^w \quad (48)$$

$$\mathbf{g}^b = \begin{bmatrix} -g \sin \theta \\ g \sin \phi \cos \theta \\ g \cos \phi \cos \theta \end{bmatrix} \quad (49)$$

where F_i is the thrust magnitude of the i th propeller, and $\mathbf{g}$ is the gravitational acceleration vector in the inertial frame.

Euler's equations for the end components are expressed as:

$$J_p \dot{\omega}_{pi}^p + \omega_{pi}^p \times (J_p \omega_{pi}^p) = \tau_{s,pi}^p + \tau_{p,pi}^p + \tau_{r,pi}^p \quad (50)$$

$$J_w \dot{\omega}_{wi}^w + \omega_{wi}^w \times (J_w \omega_{wi}^w) = \tau_{s,wi}^w + \tau_{a,wi}^w + \tau_{r,wi}^w \quad (51)$$

Since Euler's equations are established in the end-effector frame, the angular velocity and angular acceleration terms on the left-hand side of the equations are obtained by left-multiplying their expressions in the airframe frame by the corresponding attitude matrix $\mathbf{R}_{b,pi}$ or $\mathbf{R}_{b,wi}$. On the right-hand side of the equations, $\tau_{s,pi}^p$ and $\tau_{s,wi}^w$ denote the total constraint torques exerted by the airframe on the end components, and they are the unknowns to be solved in this step. $\tau_{a,wi}^w$ has been given in Eq. (28). $\tau_{p,pi}^p$ is the driving torque of the propeller, which can be expressed as:

$$\tau_{p,pi}^p = [0, 0, (-1)^i Q_i]^T \quad (52)$$

where Q_i is the magnitude of the counter-torque. Since the two propellers rotate in opposite directions, with the left propeller spinning counterclockwise about the negative direction of the Z_{pi} -axis, a negative value must be assigned to it. This design is intended to offset the counter-torque of the two propellers as much as possible, and in cruise mode, the coupling torque from the yaw axis to the roll axis can act as a positive auxiliary effect.

$\tau_{r,pi}^p$ and $\tau_{r,wi}^w$ are the torques caused by the misalignment between the point of application of the constraint force and the center of mass of the end component, which can be expressed respectively as:

$$\tau_{r,pi}^p = \mathbf{r}_{mpi,mpi}^p \times \mathbf{F}_{r,pi}^p \quad (53)$$

$$\tau_{r,wi}^w = \mathbf{r}_{mwi,rwi}^w \times \mathbf{F}_{r,wi}^w \quad (54)$$

where all constraint forces must first be transformed into the end-effector frame for representation, which can be achieved by left-multiplying the corresponding attitude matrix $\mathbf{R}_{pi,b}$ or $\mathbf{R}_{wi,b}$. $\mathbf{r}_{mpi,mpi}^p$ and $\mathbf{r}_{mwi,rwi}^w$ are the position vectors pointing from the center of mass of the

end component to the equivalent point of application of the force. In fact, the points of application of the forces can be chosen arbitrarily; this will not affect the final calculation results as long as they lie on the joint hinge axes and remain consistent throughout the derivation process. Therefore, the points of application of the forces can be directly set as the origins of their respective frames, which is expressed as:

$$r_{mpi,rpi}^p = -r_{opi,mpi}^p \quad (55)$$

$$r_{mwi,rwi}^w = -r_{owi,mwi}^w \quad (56)$$

Step6: Body Newton-Euler equations

The Newton-Euler equations of the airframe are expressed in the airframe frame. It is therefore necessary first to transform τ_{spi}^{pi} and $\tau_{s,wi}^{wi}$ into the airframe frame, which can be achieved by left-multiplying them by $R_{pi,b}$ and $R_{wi,b}$ respectively to obtain τ_{spi}^b and $\tau_{s,wi}^b$. The resultant external force acting on the airframe is defined as F_b^b , and the resultant external torque as τ_b^b , which can be expressed respectively as:

$$F_b^b = F_p^b + F_w^b + m_b g^b \quad (57)$$

$$\tau_b^b = \tau_p^b + \tau_w^b \quad (58)$$

where:

$$F_p^b = -\sum F_{r,pi}^b \quad (59)$$

$$F_w^b = -\sum F_{r,wi}^b \quad (60)$$

$$\tau_p^b = -\sum \tau_{spi}^b - \sum (r_{ob,opi}^b \times F_{r,pi}^b) \quad (61)$$

$$\tau_w^b = -\sum \tau_{s,wi}^b - \sum (r_{ob,owi}^b \times F_{r,wi}^b) \quad (62)$$

Applying the Newton-Euler equations to the airframe, we obtain its linear acceleration and angular acceleration as:

$$\dot{v}_b^b = m_b^{-1} F_b^b - \omega_b^b \times v_b^b \quad (63)$$

$$\dot{\omega}_b^b = J_b^{-1} [\tau_b^b - \omega_b^b \times (J_b \omega_b^b)] \quad (64)$$

At this point, the derivation of the relevant equations for the multi-body dynamic model is completed. The simulation model is established in the Matlab Simulink environment, aiming to provide an important reference for guiding the controller design of the actual prototype through qualitative analysis of the PW-TUAV system. Next, the open-loop response characteristics of the PW-TUAV system will be further analyzed based on this model.

3.3. Response characteristics analysis

In the PW-TUAV prototype, the vectoring rotor mechanism serves as the primary actuator. Theoretically, the vectoring rotor mechanism alone is capable of realizing all the attitude control required for the PW-TUAV during VTOL, transition, and cruise mode. However, the non-minimum phase characteristics and the small moment arm of the pitch axis lead to weak anti-disturbance capability of the aircraft. Therefore, the phase-shifting wing mechanism is incorporated as an auxiliary actuator in this study. In this section, the actual response models of the vectoring rotor mechanism and the phase-shifting wing mechanism will be established separately, followed by an analysis of their open-loop response characteristics, so as to more intuitively demonstrate their control mechanisms.

3.3.1. Simulation modeling of vectoring rotors

The design architecture of the simulation model for the vectoring rotor mechanism is shown in Fig. 4. In the entire simulation system, the

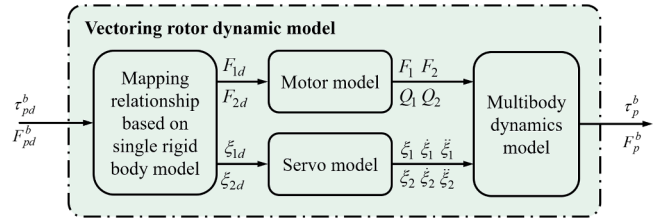


Fig. 4. Details of simulation model establishment and variable transfer relationship in the entire system for the vectoring rotor mechanism.

role of the vectoring rotor dynamic model is to take the desired propeller-generated force and torque τ_{pd}^b and F_{pd}^b output by the controller as input signals, and then output the actually responsive force and torque τ_p^b and F_p^b to the airframe part. The vectoring rotor dynamic model internally consists of three functional modules.

The detailed derivation is as follows.

First, it is necessary to convert the desired force and torque signals into the desired propeller thrust signals and desired thrust vector direction signals. This mapping relationship can be expressed analytically by establishing a single rigid-body model. In the single rigid-body model, to facilitate the expression of the control force and torque of the vectoring rotor mechanism, the control variables can be defined as:

$$\begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \end{bmatrix} = \begin{bmatrix} F_1 \sin \xi_1 + F_2 \sin \xi_2 \\ F_1 \sin \xi_1 - F_2 \sin \xi_2 \\ F_1 \cos \xi_1 + F_2 \cos \xi_2 \\ F_1 \cos \xi_1 - F_2 \cos \xi_2 \end{bmatrix} \quad (65)$$

In this way, the control force and torque generated by the vectoring rotor mechanism can be expressed respectively as:

$$F_p^b = U_3 \quad (66)$$

$$\tau_p^b = \text{diag}\{L_{yp}, L_{xp}, -L_{yp}\} \begin{bmatrix} U_4 \\ U_1 \\ U_2 \end{bmatrix} \quad (67)$$

Based on its inverse mapping relationship, the expression forms of each desired signal can be obtained as:

$$\begin{bmatrix} U_{1d} \\ U_{2d} \\ U_{3d} \\ U_{4d} \end{bmatrix} = \begin{bmatrix} \tau_p^b(2) & -\tau_p^b(3) & F_{pd}^b & \tau_p^b(1) \end{bmatrix}^T \begin{bmatrix} L_{xp} \\ L_{yp} \\ L_{yp} \\ L_{yp} \end{bmatrix} \quad (68)$$

$$\begin{bmatrix} \xi_{1d} \\ \xi_{2d} \\ F_{1d} \\ F_{2d} \end{bmatrix} = \begin{bmatrix} \text{atan}\left(\frac{U_{1d} + U_{2d}}{U_{3d} + U_{4d}}\right) \\ \text{atan}\left(\frac{U_{1d} - U_{2d}}{U_{3d} - U_{4d}}\right) \\ \frac{U_{3d} + U_{4d}}{2\cos \xi_{1d}} \\ \frac{U_{3d} - U_{4d}}{2\cos \xi_{2d}} \end{bmatrix} \quad (69)$$

Then, the desired force signals are fed into the motor model, and the desired torque signals are fed into the servo model. There are two reasons for establishing the motor model and servo model in the simulation environment: on the one hand, it is to simulate the actual delayed response characteristics; on the other hand, it is to obtain the angular velocity signals and angular acceleration signals required by the multi-body dynamic model. Since the motor model does not need to output

derivative signals with respect to time, and the overdamping characteristic of the propeller makes the system less prone to overshoot, it can be simplified as a first-order system that only takes the time-delay characteristic into account. In contrast, the servo model is underdamped and needs to output first-order and second-order derivatives with respect to time, so it is simplified as a second-order system.

The motor model also includes a mapping relationship between the desired thrust F_{id} and the desired rotational speed n_{ipd} of the propeller:

$$n_{ipd} = \left(\frac{F_{id}}{\rho_{air} D_p^4 C_{pt}} \right)^{\frac{1}{2}} \quad (70)$$

where D_p is the diameter of the propeller. The PW-TUAV prototype uses 5-inch propellers, so D_p is approximately 0.127 m. C_{pt} is the thrust coefficient of the propeller, which is about 0.176 according to the official datasheet.

The propeller speed model can be simplified as a standard first-order

system, and its transfer function is given by:

$$\frac{n_p(s)}{n_{pd}(s)} = \frac{1}{0.05s + 1} \quad (71)$$

where the time constant is 0.05, a typical value for the combination of a high-kV motor and a small-diameter propeller.

The actual propeller thrust and counter-torque can be calculated from the actual rotational speed using the following equations:

$$F_i = \rho_{air} n_p^2 D_p^4 C_{pt} \quad (72)$$

$$Q_i = \rho_{air} n_p^2 D_p^5 C_{pq} \quad (73)$$

According to the official datasheet, the value of C_{pq} is approximately 0.025.

The servo model can be directly simplified as a standard second-order system. Since it is also necessary to observe the first-order and

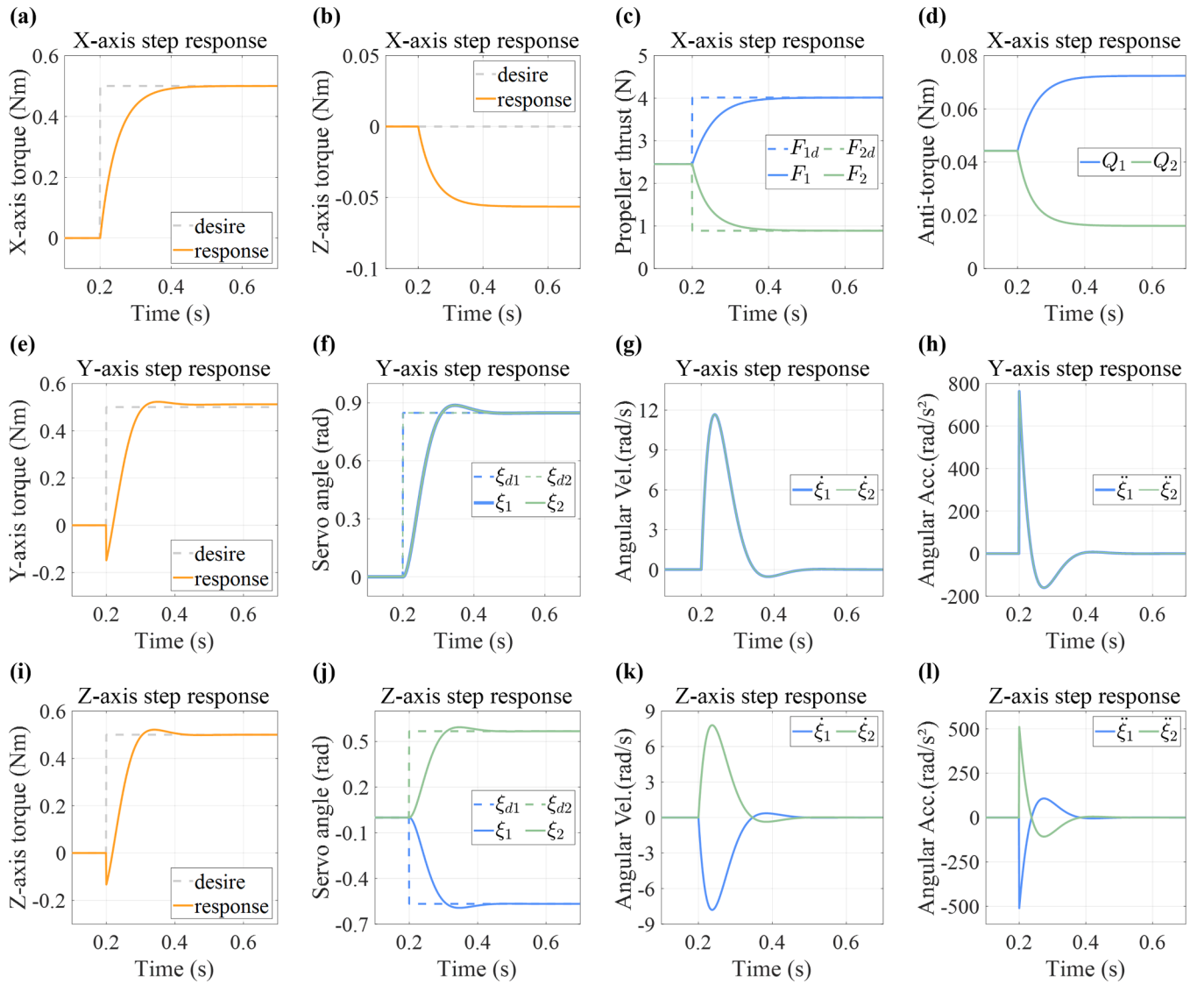


Fig. 5. Response characteristics of the vectoring rotor mechanism to desired torque step signals. The desired torque values in the first row are set to [0.5; 0; 0], those in the second row to [0; 0.5; 0], and those in the third row to [0; 0; 0.5]. (a) Response curve to the X-axis step signal. (b) Z-axis torque response induced by the X-axis step signal. (c) Variation trend of propeller thrust under the X-axis step signal. (d) Variation trend of propeller counter-torque under the X-axis step signal. (e) Response curve to the Y-axis step signal. (f) Variation trend of servo angle under the Y-axis step signal. (g) Variation trend of servo angular velocity under the Y-axis step signal. (h) Variation trend of servo angular acceleration under the Y-axis step signal. (i) Response curve to the Z-axis step signal. (j) Variation trend of servo angle under the Z-axis step signal. (k) Variation trend of servo angular velocity under the Z-axis step signal. (l) Variation trend of servo angular acceleration under the Z-axis step signal.

second-order derivatives with respect to time, the state-space model is established as follows:

$$\begin{bmatrix} \dot{\xi} \\ \ddot{\xi} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{bmatrix} \xi \\ \dot{\xi} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} \xi_d \quad (74)$$

$$\begin{bmatrix} \dot{\xi} \\ \ddot{\xi} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{bmatrix} \xi \\ \dot{\xi} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \omega_n^2 \end{bmatrix} \xi_d \quad (75)$$

This second-order system takes the desired tilt angle ξ_d as input, and outputs the actual tilt angle ξ along with its corresponding angular velocity $\dot{\xi}$ and angular acceleration $\ddot{\xi}$. Based on the official servo test data, the natural frequency ω_n of the drive servo for the vectoring rotor mechanism can be set to 30, and the damping ratio ζ can be set to 0.7, with a rise time of approximately 0.08 s in response to a step signal.

3.3.2. Control mechanism of vectoring rotors

Fig. 5 illustrates the response characteristics of the vectoring rotor mechanism under step signals of different axes. Since no controller is involved in this analysis process, the airframe can be regarded as stationary, and only how the influence of the vectoring rotor is expressed in the airframe frame is considered, which serves to elaborate on the control principle diagram in Fig. 2(e). The initial propeller thrust is equal to the gravity of the airframe. It can be seen from Fig. 5(a)–(d) that the vectoring rotor mechanism obtains the X-axis torque in the airframe frame by means of differential propeller rotational speeds, which manifests as roll torque during the hover mode. However, the change in propeller rotational speed is inevitably accompanied by a change in counter-torque, thereby inducing a coupling torque around the Z-axis. When the total propeller thrust is close to the gravity of the prototype, the magnitude of this undesired coupling torque is approximately one-tenth of the X-axis response torque, which may require additional compensation in the controller for applications with high control accuracy requirements. Fig. 5(e) shows the response characteristics of the vectoring rotor mechanism to the Y-axis step signal, which manifests as pitch torque throughout all flight phases. It is evident that the vectoring rotor mechanism exhibits obvious non-minimum phase characteristics, generating a torque opposite to the desired direction immediately after the step signal is applied. The multi-body dynamic model can clearly characterize this phenomenon. Fig. 5(f)–(h) display the variation trends of the servo angle, angular velocity, and angular acceleration corresponding to Fig. 5(e); the variation trends of the left and right servos are identical, with their curves overlapping. Fig. 5(i)–(l) present the response characteristics of the vectoring rotor mechanism to the Z-axis step signal, which also exhibits non-minimum phase characteristics. This non-minimum phase characteristic limits the controller gain from being excessively large, directly restricting its performance upper limit. In addition, a comparison between Fig. 5(f) and Fig. 5(j) indicates that the vectoring rotor mechanism has low efficiency in generating pitch torque, requiring a larger tilt angle of the servo. Although this issue can be alleviated by extending the installation position of the vectoring rotor mechanism along the negative direction of the airframe Z-axis, this solution has two drawbacks: on the one hand, the protruding propellers may result in an awkward appearance of the aircraft; on the other hand, it will lead to more severe non-minimum phase effects, posing stringent challenges to the controller—especially when the gyroscope noise is significant.

Overall, the non-minimum phase characteristics of the vectoring rotor mechanism about the Y-axis and Z-axis in the airframe frame limit the upper performance limit of its controller. Meanwhile, the Y-axis suffers from the issue of a short control moment arm, and the Z-axis is accompanied by coupling torque induced by the X-axis response. Therefore, this study incorporates the phase-shifting wing mechanism as

an auxiliary control component; the auxiliary control torque it generates acts exactly about the Y-axis and Z-axis of the airframe, which can compensate for the shortcomings of the vectoring rotor mechanism to a certain extent. The simulation model and response characteristics of the phase-shifting wing mechanism are presented below.

3.3.3. Simulation modeling of phase-shifting wings

The design architecture of the simulation model for the phase-shifting wing mechanism is shown in Fig. 6. The function of the entire phase-shifting wing dynamic model module is to take the desired phase angle λ_{id} as input, and then output the resultant force F_w^b and torque τ_w^b generated by the wing section to the airframe section. The interior of the phase-shifting wing dynamic model also consists of three modules, but it is slightly more complex than the vectoring rotor dynamic model.

First, the wing kinematic phase model is introduced. The input desired phase angle λ_{id} is converted into the desired crank angle Cr_{id} via the mapping relationship presented in Fig. 3(e). Its forward mapping formula is shown in Eq. (99). Directly deriving its analytical inverse mapping formula may be difficult and result in unnecessary waste of computational resources; therefore, the polynomial fitting method is adopted for fast calculation. Subsequently, the desired crank angle Cr_{id} is fed into the servo model to obtain the actual responsive crank angle as well as its corresponding angular velocity and angular acceleration. The establishment method of the servo model is identical to that used in the vectoring rotor dynamic model, except that the natural frequency ω_n is set to 36 and the damping ratio to 0.75 according to the specific type of servo selected. Then, based on the previously derived Eq. (99); Eq. (101) and Eq. (102), a four-bar linkage model is established, which maps the responsive crank angle Cr_i , angular velocity $\dot{Cr}_i$ and angular acceleration $\ddot{Cr}_i$ to the responsive phase angle λ_i , angular velocity $\dot{\lambda}_i$ and angular acceleration $\ddot{\lambda}_i$ of the wing, respectively.

The second module is the wing aerodynamic model, whose function is to obtain the aerodynamic force and aerodynamic torque in the wing frame; the specific implementation scheme has been elaborated in detail in Section 3.1. Finally, a multi-body dynamic model is established based on the scheme presented in Section 3.2, which outputs the resultant force F_w^b and resultant torque τ_w^b generated by the wings on both sides to the airframe section.

To highlight the significance of the phase-shifting function design, a quantified observation model is additionally established in the simulation environment. In this model, the wings are fixed in position, consistent with the configuration of conventional tail-sitters. Then, through a separate set of wing aerodynamic model and multi-body dynamic model, the resultant force F_{w0}^b and resultant torque τ_{w0}^b of the wings under the fixed-wing condition are obtained. The differential values ΔF_w^b and $\Delta \tau_w^b$ are derived by subtraction, which are used to quantitatively observe the additional auxiliary control variables generated by the phase-shifting wing mechanism in comparison with the fixed-wing design. The control mechanism of the phase-shifting wing mechanism is analyzed based on these data.

3.3.4. Control mechanism of phase-shifting wings

Since this study focuses primarily on attitude control, only the torque is analyzed. Fig. 7 illustrates the response characteristics of the additional auxiliary control torque generated by the phase-shifting wing mechanism compared with the fixed-wing design, using the right wing as an example. Given that part of this auxiliary control torque is generated by reducing the effective projected area to mitigate the impact of external wind disturbances, a uniform external wind disturbance with a velocity of 5 m/s along the negative direction of the X-axis in the airframe frame is applied in the test conditions.

Fig. 7(a) shows the response characteristics of the wing phase angle to the DV, where a step signal with a DV of 45° is applied. Since the wing is not expected to change its phase angle at a high speed to avoid excessive rotational damping force, a low-pass filter needs to be added to

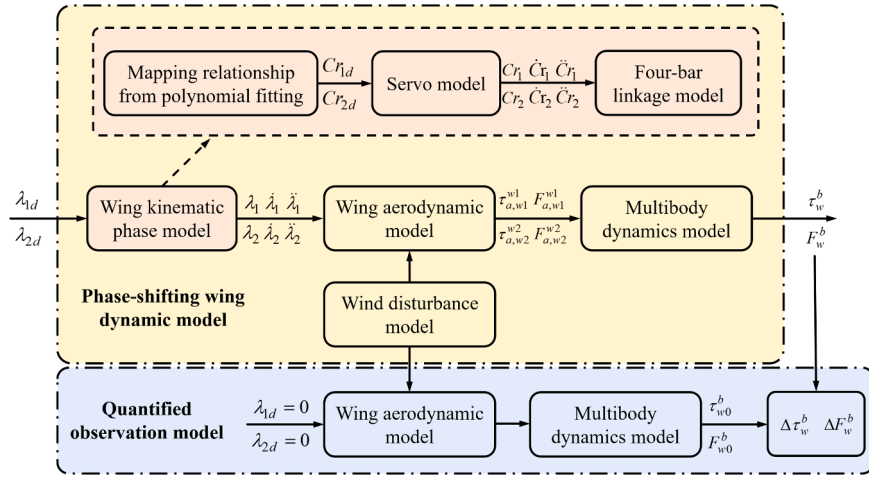


Fig. 6. Details of simulation model establishment and variable transfer relationship in the entire system for the phase-shifting wing mechanism.

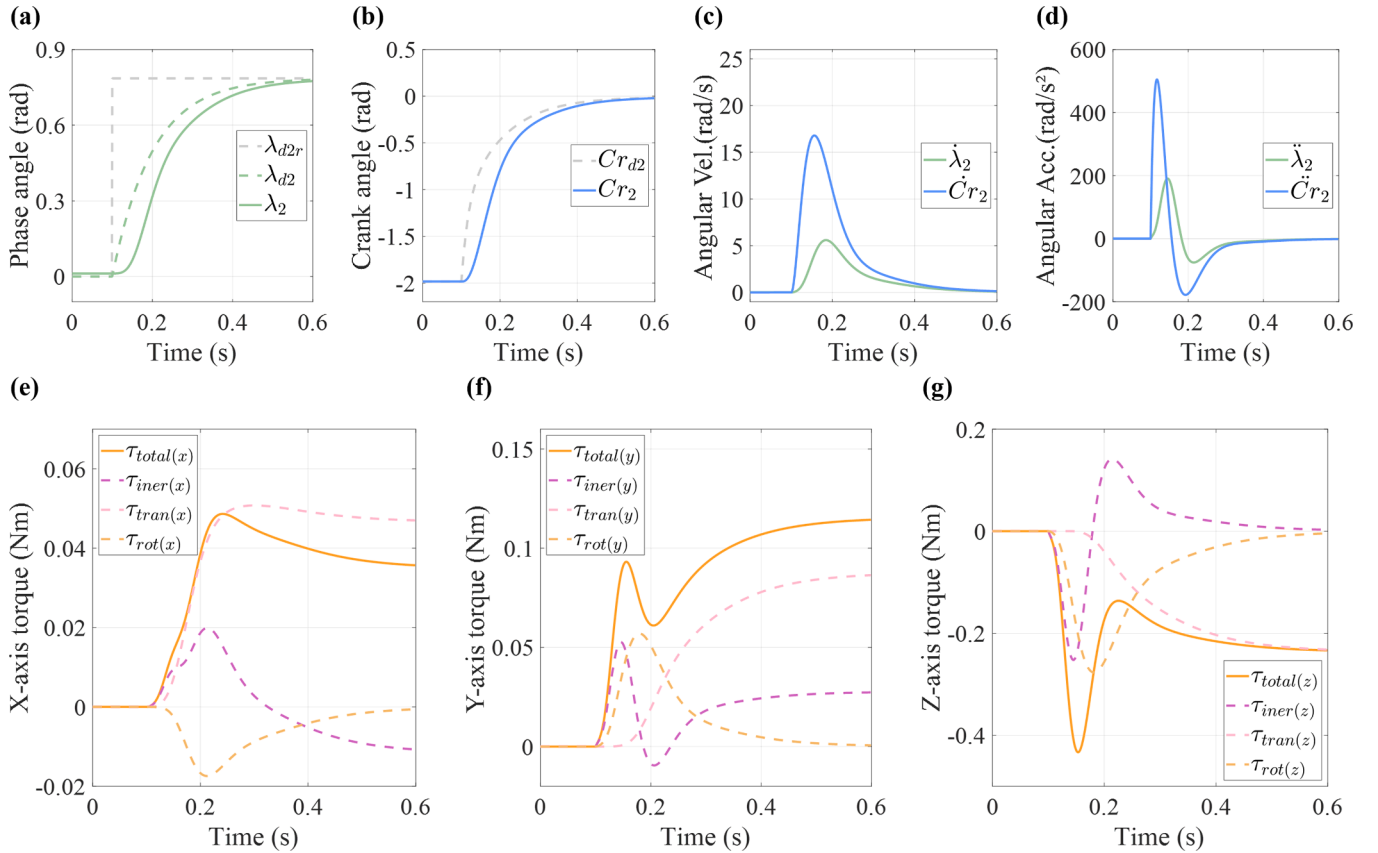


Fig. 7. Auxiliary control mechanism of the phase-shifting wing mechanism. (a) Response characteristics of the wing phase angle to the desired value (DV). (b) Corresponding crank angle. (c) Variation trend of angular velocity. (d) Variation trend of angular acceleration. (e) Auxiliary control torque about the X-axis. (f) Auxiliary control torque about the Y-axis. (g) Auxiliary control torque about the Z-axis.

the desired phase angle signal. In the Simulink environment, this is implemented by adding a first-order system with a time constant of 0.1, which converts the reference signal λ_{d2r} output by the controller into a smooth desired signal λ_{d2} . Fig. 7(b) presents the corresponding response characteristics of the crank angle, i.e., the actual travel angle of the drive servo. Fig. 7(c)–(d) show the corresponding variation trends of angular velocity and angular acceleration. Fig. 7(e)–(g) intuitively demonstrate the auxiliary control torque induced by the wing's phase angle adjustment. The response about the X-axis is not the desired control torque but is induced by the torque coupling phenomenon of other axes, with a

magnitude much smaller than those of the other axes. The focus is placed on analyzing the torque response characteristics about the Y-axis and Z-axis. In Fig. 7(f) and Fig. 7(g), the main contributing factors to the torque are solved separately and presented in the form of curves, namely the inertial term, translational damping attenuation term, and rotational damping term. Coincidentally, the torques generated by these influencing factors are in the same direction, and there is no non-minimum phase characteristic unlike that of the vectoring rotor mechanism, which is highly favorable for controller design. Fig. 7(f) and Fig. 7(g) clearly show that: the inertial term determines the initial rising rate of

the auxiliary control torque; the rotational damping term promotes the rising rate in the middle stage but may cause slight oscillations when the desired phase angle changes excessively; the translational damping attenuation term has a slow initial rising rate but mainly determines the final steady-state value of the auxiliary control torque. The inertial term incorporates gravity, thus also generating an auxiliary control torque about the Y-axis, but its proportion is relatively small.

Overall, the response characteristics about the Y-axis indicate the following: when the aircraft is subjected to head-on wind during the hover mode, the aerodynamic center is located at the half-chord position, which is far from the aircraft's center of gravity, thus generating a strong nose-down torque. Adjusting the wing phase angle can generate a nose-up torque through multiple factors to counteract the effect of wind disturbance. Since the Y-axis torques generated by the wings on both sides are in the same direction, adjusting the phase angles of both wings simultaneously can double this control torque. The same principle applies to the response characteristics about the Z-axis: when the aircraft is affected by a unilateral wind gust, adjusting the phase angle of the wing on one side can generate a reverse auxiliary control torque, thereby significantly attenuating the impact of external disturbances. However, due to the large moment arm of the rotational damping force about the Z-axis, a more obvious overshoot phenomenon occurs in the middle stage of the response. Therefore, when designing the actual controller, it is necessary to constrain the difference in rotational angular velocities between the wings on both sides to avoid excessive values. The above analysis shows that the auxiliary control mechanism of the phase-shifting wing mechanism mainly relies on its steady-state characteristics. Based on this feature, this study proposes an integral-driven multi-actuator SC for attitude control, which dynamically distributes the control variables between the phase-shifting wing mechanism and the vectoring rotor mechanism.

4. Controller design

Based on the established dynamic model in the preceding section, this section presents a design method of SC for the PW-TUAV configuration.

Natural biological birds possess sophisticated wing-tail synergistic

control capabilities [64], but this is in fact closely related to their airflow perception capabilities. Currently, however, limited by the technical constraint of the difficulty in miniaturizing airborne sensors, it is difficult for small UAVs to deploy micro pressure sensors on the wings that satisfy both sufficient lightweight and high measurement accuracy, posing great challenges to the design of SC. This study aims to demonstrate that the designed PW-TUAV prototype can still effectively achieve synergistic control between the phase-shifting wing mechanism and the vectoring rotor mechanism even when equipped only with basic sensors such as an inertial measurement unit (IMU) and an optical flow sensor (OFS), thereby significantly improving the aircraft's anti-disturbance performance during VTOL and transition phases. This is the case even though the performance potential of the phase-shifting wing mechanism cannot be exploited more efficiently due to limited perception capabilities. This study provides a more easily implementable design scheme for the SC that is highly favorable for the early-stage research and exploration of such aircraft. The control architecture for the PW-TUAV is presented in Fig. 8.

First, the design mechanism of the overall controller is outlined. In the hover mode, the controller adopts a double closed-loop structure, with an outer position control loop and an inner attitude control loop. Among these, the altitude data is fed into the altitude controller for independent processing, which is a simple PID controller that directly yields the desired propeller thrust F_{pd}^b in the airframe frame. The horizontal position data is input to the horizontal position controller, which is also a simple PID controller, and outputs the desired roll angle and pitch angle to the attitude controller. In the transition phase, however, the desired attitude transition trajectory is preset, so only the attitude control loop is involved. The attitude controller implements synergistic control of the phase-shifting wing mechanism and the vectoring rotor mechanism, and its outputs consist of two parts: the desired main control torque τ_{pd}^b generated by the vectoring rotor mechanism and the desired wing phase angle λ_{id} . To minimize the overshoot and oscillation issues mentioned at the end of Section 3.3.4, a low-pass filtering process is applied to the λ_{id} signal. These desired control variables are input into the vectoring rotor dynamic model and phase-shifting wing dynamic model established in Section 3.3. The actual responsive physical

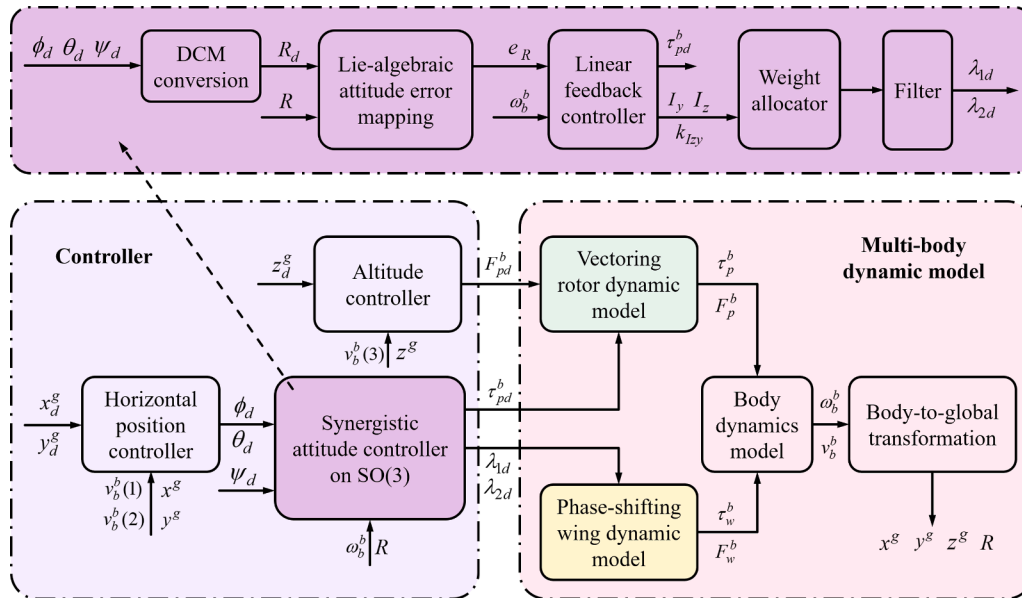


Fig. 8. Control architecture for the VTOL and transition phases of PW-TUAV. In the controller block diagram, all desired signals are input via left-side incoming arrows, while all feedback state signals are input via bottom-side incoming arrows. The controller is designed identically for both the simulation environment and the physical prototype system. In simulation, a multi-body dynamics model is used to solve for Euler angles and ground-relative displacement. In the actual prototype, these data are derived from the fusion of IMU and OFS measurements performed by the state estimator. Further details on practical deployment are provided in Fig. 12.

quantities are then derived, integrated into the airframe dynamic model, and finally converted via coordinate transformation to obtain the pose data of the aircraft in the ground coordinate system. The more complex part of the work lies in the design of the SC, which mainly includes two aspects: first, a full-attitude controller for the vectoring rotor mechanism, and second, a multi-actuator dynamic allocation scheme.

Full-attitude control is a common control challenge faced by tail-sitter UAVs. When such aircraft transition from VTOL to cruise mode, the pitch angle undergoes a large-angle maneuver of nearly 90° , which inevitably leads to the gimbal lock problem of Euler angles. Some studies adopt a ZXY-based rotation sequence [27] or a hybrid attitude representation method combining quaternions and Euler angles [28] to address this issue. This study employs a method based on the Special Orthogonal Group in three dimensions (denoted as SO(3)) [65]. The control law is designed on a smooth manifold structure, enabling a global singularity-free attitude representation, which is more conducive to the future development of aerobatic maneuvers for the aircraft.

Define the desired rotation attitude matrix as R_d , with the rotation sequence specified as ZYX; thus, R_d can be expressed as:

$$R_d = R_z(\psi_d)R_y(\theta_d)R_x(\phi_d) \quad (76)$$

where ϕ_d , θ_d , and ψ_d denote the desired roll angle, pitch angle, and yaw angle, respectively. This process is referred to as direction cosine matrix (DCM) conversion. In the feedback loop of the controller, the current rotation attitude matrix R is directly observed without the need to derive Euler angle data from it. In the body-to-global transformation module, the real-time rotation attitude matrix R is updated based on the differential equation and exponential mapping relationship of the rotation matrix, and its expression is given by:

$$\dot{R}(t) = R(t) \cdot \text{sk}(\omega_b^b) \quad (77)$$

$$R(t + \Delta t) = R(t) \cdot \exp[\text{sk}(\omega_b^b) \Delta t] \quad (78)$$

where the $\text{sk}(\cdot)$ operator denotes the skew-symmetric mapping [56], which is the inverse mapping of the $(\cdot)^\vee$ operator.

According to Rodrigues' formula, the exponential mapping of a matrix can be expressed as:

$$\exp[\text{sk}(\omega_b^b) \Delta t] = I + \frac{\sin(\|\omega_b^b\| \Delta t)}{\|\omega_b^b\|} \text{sk}(\omega_b^b) + \frac{1 - \cos(\|\omega_b^b\| \Delta t)}{\|\omega_b^b\|^2} [\text{sk}(\omega_b^b)]^2 \quad (79)$$

The rotation error is expressed in the form of a matrix:

$$R_e = R_d^T R \quad (80)$$

Via logarithmic mapping, the rotation error matrix can be mapped to the Lie algebra:

$$e_R = [\log(R_e)]^\vee \quad (81)$$

Based on the inverse Rodrigues' formula, the logarithmic mapping of a matrix can be expressed as:

$$\log(R_e) = \frac{\kappa}{2\sin\kappa} (R_e - R_e^T) \quad (82)$$

where κ is the rotation angle, calculated by the formula:

$$\kappa = \arccos\left(\frac{\text{tr}(R_e) - 1}{2}\right) \quad (83)$$

Then, the rotation error in the form of Lie algebra is fed into the linear feedback controller module. A simple yet effective approach [56, 62] is to adopt the structure of a PID controller, with parameter tuning assisted by the particle swarm optimization (PSO) algorithm, so as to eliminate the influence caused by manual parameter tuning. This is highly necessary because the three rotational axes correspond to three PID controllers and nine control parameters, making manual parameter

tuning extremely challenging. The implementation procedure and parameter configuration of PSO are detailed in Appendix D.

In the linear feedback controller module, the proportions of the integral term, denoted as I_y and I_z , in the controller outputs for the y-axis and z-axis are recorded, and the proportionality coefficient k_{IZY} between them is calculated, which is used to design the multi-actuator dynamic allocation scheme. Due to the limitations of onboard sensors and computational resources, some more advanced optimal control strategies are difficult to implement in the PW-TUAV system. This study proposes an integral-driven multi-actuator SC, which can be implemented with only an IMU sensor and does not require substantial computational resources. Of course, this method has certain limitations: it cannot distinguish whether the disturbance is caused by external wind gust variations or other factors. However, UAVs at the current stage generally perform tasks in open spaces and are not temporarily involved in complex behaviors such as human-UAV interaction. Therefore, the proposed SC has high applicability at the current stage.

The anti-disturbance mechanism of the SC is implemented via a weight allocator. First, it should be clarified that since wind disturbance conditions are unknown, the designed SC is unilateral, which aims to mitigate external disturbances by means of dynamically adjusted wing phase angles. In contrast, the desired control torque τ_{pd}^b of the vectoring rotor mechanism will not actively reduce its weight share because of this. This does not mean that the weight share of τ_{pd}^b will never decrease; instead, the decrease is passive, occurring as a consequence of the reduction in feedback error. Such a design renders the controller output smoother, as the integral term will not change abruptly.

As can be seen from Fig. 2(e), adjusting the phase angles of the wings on both sides simultaneously can assist in the control of the pitch axis, while differential adjustment can assist in the control of the yaw axis in the hover mode. First, it is necessary to calculate the weight ratios of these two actions separately. When the absolute value of the proportion of the integral term is <0.2 , the long-term continuous external wind disturbance is considered slight, and the vectoring rotor mechanism is sufficient to cope with it. In this case, the phase-shifting wing mechanism is not required to participate, and its weight is set to 0. When the absolute value of the proportion of the integral term is greater than 0.2, the auxiliary control of the phase-shifting wing mechanism is activated, and a smooth cubic Bezier curve is adopted to map the respective weight values. Therefore, the expression form of this piecewise function is as follows:

$$w_{I_i} = \begin{cases} 0, & |I_i| \leq 0.2 \\ \text{sgn}(I_i) \cdot B\left(\frac{|I_i| - 0.2}{0.8}\right), & 0.2 < |I_i| \leq 1 \end{cases} \quad (84)$$

$B(x) = -0.5x^3 + 1.2x^2 + 0.3x$ (85) where I_i denotes I_y , it corresponds to the weight coefficient w_{I_1} ; where I_i denotes I_z , it corresponds to the weight coefficient w_{I_2} .

Then, the weights w_{I_1} and w_{I_2} are reallocated based on k_{IZY} , and the new weight coefficients can be expressed as:

$$w_{I_1}' = w_{I_1} + (1 - w_{I_1})w_{I_2}^+ - k_{IZY}w_{I_1}w_{I_2}^- \quad (86)$$

$$w_{I_2}' = w_{I_2} + (1 - w_{I_2})w_{I_1}^- - k_{IZY}w_{I_1}w_{I_2}^+ \quad (87)$$

where:

$$w_{I_2}^+ = \max(w_{I_2}, 0) \quad (88)$$

$$w_{I_2}^- = \max(-w_{I_2}, 0) \quad (89)$$

The desired wing phase angle is finally obtained as:

$$\lambda_{id} = k_\lambda w_{I_i}' \quad (90)$$

where k_λ is the gain coefficient, set to 0.785 to balance the frontal and lateral projections of the airframe while avoiding the problem of insufficient lift during the transition phase.

5. Validation and discussion

This section will verify the effectiveness of the SC designed in the preceding section during the hover mode and transition phase of the aircraft through simulation and outdoor flight tests of the actual prototype, respectively.

5.1. Simulation validation

5.1.1. Simulation environment setup

The simulation model is established in the MATLAB Simulink environment. All involved formulas and aircraft parameters have been presented in the previous two sections, and the design architecture refers to Fig. 4; Fig. 6, and Fig. 8. All initial states of the aircraft—including position, velocity, and acceleration relative to the ground, as well as Euler angles, angular velocities, and angular accelerations of the attitude—are set to 0. The global simulation step size is set to 0.001 s, and the feedback control step size is 0.01 s (consistent with the real hardware configuration environment), with the solver selection set to automatic. The servo actuators need to be configured with corresponding performance amplitude constraints based on the performance indicators introduced in Section 2.2.

For the purpose of comparative evaluation, a deterministic discrete gust model is adopted to provide a clear and reproducible input, facilitating the assessment of the flight control system's characteristics. The gust model is established in the global coordinate system and is considered to consist of two parts:

$$\mathbf{v}_{gust}^g = \mathbf{v}_{gust,c}^g + \tilde{\mathbf{v}}_{gust,r}^g \quad (91)$$

$$\mathbf{v}_{gust,c}^g = \begin{cases} 0, & t \leq t_0 \\ \frac{K_c}{2} \left[1 - \cos\left(\frac{\pi(t-t_0)}{t_1}\right) \right], & t_0 < t < t_0 + t_1 \\ K_c, & t \geq t_0 + t_1 \end{cases} \quad (92)$$

$$\tilde{\mathbf{v}}_{gust,r}^g = K_r \mathcal{N}(0, 1) \quad (93)$$

Among these, $\mathbf{v}_{gust,c}^g$ is the main gust disturbance term established based on the 1-cosine rise curve, where K_c denotes the peak wind speed, t_0 is the moment when the gust starts to occur, and t_1 is the time required for the gust to reach its peak value. To facilitate the observation and comparison of the performance differences between different control methods, the gust is kept constant after rising to the peak value. $\tilde{\mathbf{v}}_{gust,r}^g$ is the random disturbance term, obtained via low-pass filtering of the randomly fluctuating noise sequence $\mathbf{v}_{gust,r}^g$. K_r is the gain coefficient of random fluctuation, and $\mathcal{N}(0, 1)$ represents a Gaussian white noise sequence with a mean of 0 and a variance of 1. To ensure the complete consistency of the gust model in the comparative simulation experiments, a fixed random number seed is adopted to generate reproducible pseudorandom white noise.

Flight experiments under three types of disturbance conditions were mainly carried out in the simulation environment, namely: **a)** head-on gust disturbance in hover mode, **b)** local lateral wind disturbance in hover mode, and **c)** local lateral wind disturbance during the transition phase. These three types of disturbance conditions correspond to the three flight performance requirements of such tail-sitter UAVs respectively: **a)** the capability to maintain fixed-point hovering, **b)** the capability to maintain yaw orientation, and **c)** the capability to stably transit to cruise mode.

The test of forward wind disturbance during the transition phase was not conducted, because the ability of the aircraft to maintain its original position is not assessed during the transition phase, while the attitude stability performance is of greater importance at this stage. Forward wind disturbance within the normal wind speed range will not significantly affect the attitude stability, and can even help shorten the tran-

sition time as the process of accumulating forward flight speed is accelerated. However, local wind disturbance has a very significant impact on the aircraft. This is because the aircraft itself needs to perform large-angle rotation around the y_b -axis, and if there is a disturbance around the z_b -axis at this time, it will trigger a coupling effect between the y_b -axis and z_b -axis of the aircraft, making the aircraft prone to loss of control and crash.

The disturbance conditions for each experiment in the simulation environment are set as follows:

- Set $K_c=5$, $K_r=2$, $t_0=1$ and $t_1=6$. The initial direction of the $\mathbf{v}_{gust,c}^g$ term is along the negative direction of the x_g -axis, and the initial direction of the $\mathbf{v}_{gust,r}^g$ term is along the vector direction of $(-1, -1, -1)$ in the global coordinate system. The simulation results are shown in Fig. 9.
- Set $K_c=4$ for the left wing and $K_c=0$ for the right wing, with other parameters kept unchanged. The simulation results are shown in Fig. 10.
- Set $K_c=3.33$ for the left wing, with the initial direction of the $\mathbf{v}_{gust,c}^g$ term along the vector direction of $(-1, 0.5, 0)$ in the global coordinate system to simulate the scenario where a single wing is subjected to lateral-frontal wind disturbance, while other disturbance parameters remain unchanged. A transition command is sent to the aircraft at $t = t_0$, and the simulation results are shown in Fig. 11.

Since the focus of this research is to improve the attitude disturbance rejection performance and does not involve complex three-dimensional coordinate tracking control, the preset transition trajectory is a simple and easily reproducible time-varying pitch angle trajectory. The aircraft is expected to keep the roll angle and yaw angle unchanged, and gradually change the pitch angle from 0° to -70° at a rate of $-20^\circ/\text{s}$. The same preset transition trajectory scheme was also adopted in the flight experiments of the actual prototype.

Two controllers were compared in the simulation experiments: one is the SC proposed in this study, and the other is the vectoring rotor controller (VRC). VRC refers to a controller that adopts fixed wings without involving wings in disturbance rejection control, and only regulates the magnitude and direction of the thrust of vectoring rotors. It represents a relatively novel technique in the domain of tail-sitter UAVs [28]. A comparative analysis of the performance discrepancies between SC and VRC implemented on the same prototype is performed to verify the disturbance rejection performance enhancement of the proposed SC scheme.

5.1.2. Simulation result analysis

The analysis of all simulation experimental results is organized below around the core research objectives of attitude response characteristics, control effort efficiency, position holding capability, and limit disturbance rejection performance. The intuitive three-dimensional representation of the real-time gust model for each simulation experiment is presented in the corresponding subplot (a). In addition, although the controller is designed directly based on the rotation matrix and Lie algebra, these concepts are overly abstract and difficult to comprehend. For this reason, they are still converted into the form of Euler angles for observation to enable intuitive visualization.

Analysis of attitude response characteristics:

In the presence of intense head-on wind disturbance, it can be seen from Fig. 9(c) that the position control loop of the aircraft outputs the desired pitch angle, causing the aircraft to tilt and thereby adjust the direction of propeller thrust to counteract the external wind disturbance, thus maintaining the horizontal position of the aircraft. Under the same head-on wind disturbance conditions, the use of the SC compared with the VRC can significantly reduce the absolute value of the desired pitch angle. This indicates that the impact of head-on wind disturbance is passively mitigated, and the control efficiency is improved. Meanwhile, as shown in Fig. 9(b) and Fig. 9(d), the adoption of the SC leads to

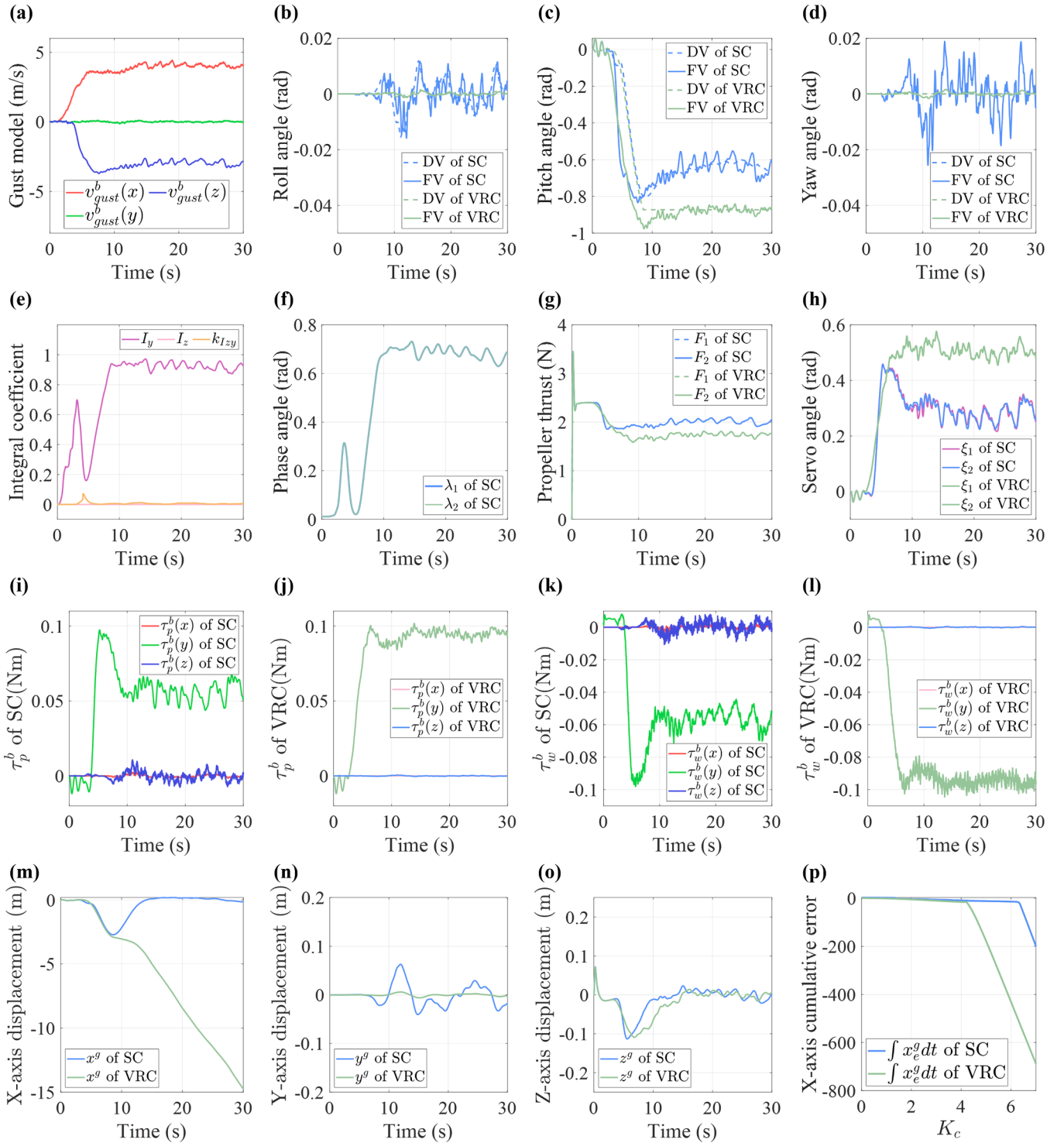


Fig. 9. Simulation results of hover mode under head-on wind disturbance conditions. (a) Time-varying values of the gust model in the airframe frame. (b)–(d) Euler angle feedback data under different controllers, where FV is the abbreviation for feedback value. (e) Time-varying curve of the integral term proportion. (f)–(h) Output responses of each actuator. (i)–(l) Torques generated by each actuator. (m)–(o) Position feedback data of the aircraft. (p) Comparison of limit head-on gust rejection capabilities of different controllers.

larger-amplitude oscillations in the aircraft's roll angle and pitch angle under the disturbance of the randomly fluctuating $v_{gust,r}^g$. Although this is an adverse effect, it is considered acceptable due to the small oscillation amplitude.

When the aircraft is subjected to local wind disturbance, the advantage of reducing the desired pitch angle becomes more pronounced, which is fully demonstrated in Fig. 10(c). At the same time,

Fig. 10(b) shows that the adoption of the SC will induce an additional desired roll angle. This is because the differentially adjusted wings lead to an increase in the lateral component of the force, and a corresponding roll angle must be generated to maintain the fixed position of the aircraft. However, this phenomenon is acceptable for two reasons. On the one hand, the increased torque demand around the x_b axis is far smaller than the reduced torque demand around the other two axes. On

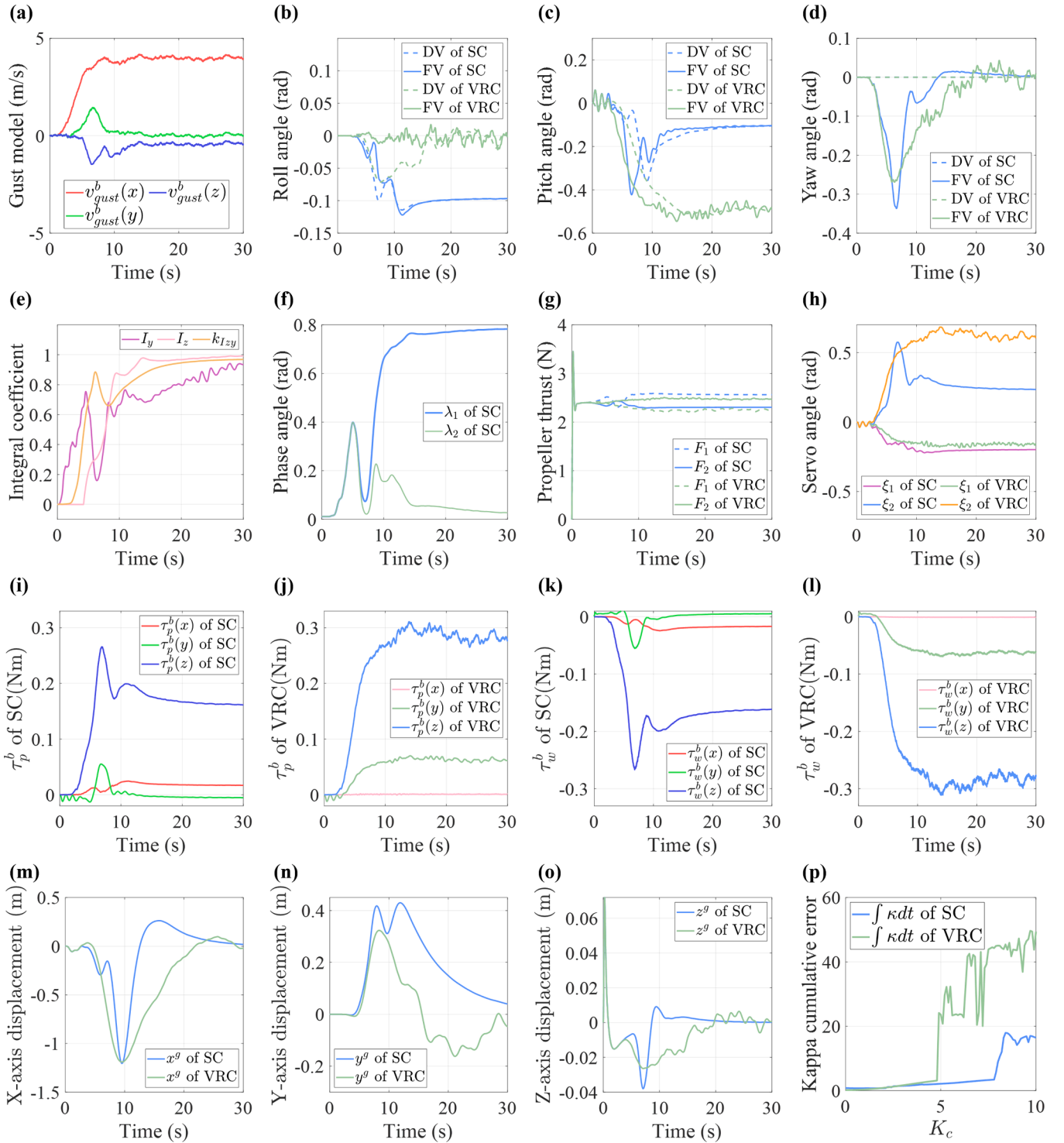


Fig. 10. Simulation results of hover mode under local wind disturbance conditions. (a)–(o) Definitions of each subfigure are consistent with those in Fig. 9. (p) The cumulative value of the axis angle is used instead to describe the disturbance rejection performance of the aircraft.

the other hand, as analyzed in Section 3.3.2, the response characteristic of the aircraft around the x_b -axis presents a minimum-phase system, while those around the y_b -axis and z_b -axis both exhibit non-minimum-phase system characteristics.

When the aircraft is subjected to local lateral wind disturbance during the execution of the transition phase, it can be seen from Fig. 11 (b) to Fig. 11(d) that the tracking control accuracy of the roll angle and yaw angle is severely affected. However, compared with the VRC, the adoption of the SC enables the attitude angles to stabilize more rapidly

with a smaller oscillation amplitude.

Analysis of control effort efficiency:

In the test experiment of head-on wind disturbance in hover mode, the advantage of adopting the SC is clearly demonstrated in Fig. 9(h), where the servo rotation angle of the VRC alone has approached the upper stroke limit. This phenomenon can be explained by Fig. 9(i) and Fig. 9(j). It can be seen that the adoption of the SC can reduce the control torque output of the vectoring rotor mechanism. This is because the effective projected area of the wing is reduced under the control of the

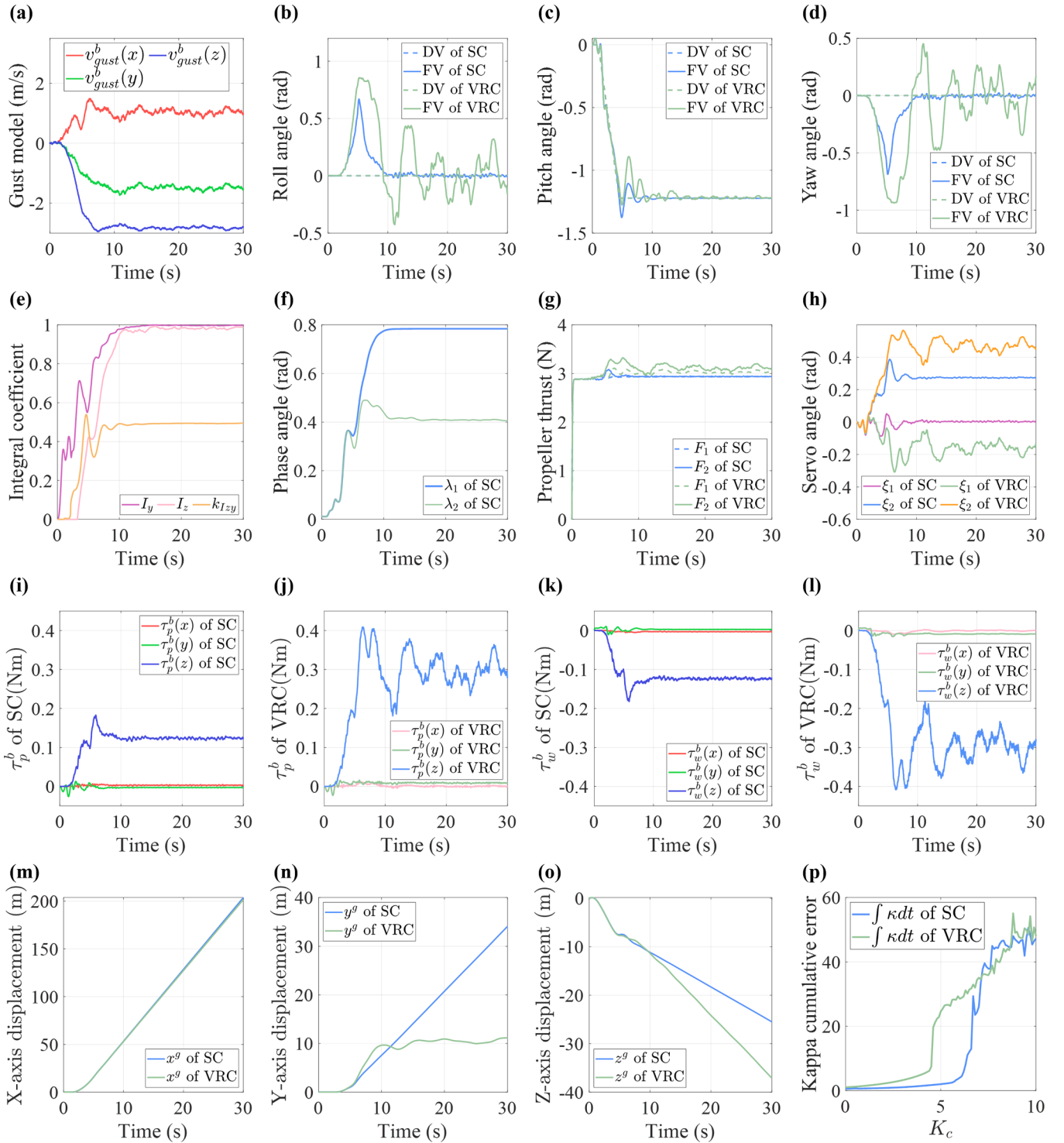


Fig. 11. Simulation results of transition phase under local wind disturbance conditions. Definitions of all subfigures are completely consistent with those in Fig. 10.

phase-shifting mechanism, thereby passively attenuating the impact of external wind disturbance. By comparing Fig. 9(i) and Fig. 9(j), it can be found that the adoption of the SC can significantly reduce the undesirable torque generated by the wing around the y_b -axis of the body frame.

In the test experiment of local wind disturbance in hover mode, it can be seen from Fig. 10(f) that the SC differentially adjusts the phase angles of the wings on both sides to reduce the effective projected area of the disturbed side, thereby mitigating the impact caused by the disturbance. This is particularly reflected in Fig. 10(i) to Fig. 10(l). It can be observed that the use of the SC can significantly reduce the control torques around

the y_b -axis and z_b -axis of the airframe frame compared with the VRC, thus improving the control efficiency.

In the test experiment of transition phase, it can be seen from Fig. 11(g) to Fig. 11(h) that the adoption of the SC yields smoother controller commands than the use of the VRC alone. It can also be observed from Fig. 11(i) to Fig. 11(l) that the SC can significantly reduce the control torque around the z_b -axis of the body frame, thereby improving the control efficiency.

Analysis of position holding capability: Fig. 9(m) to Fig. 9(o) present the displacement feedback data of the aircraft under head-on

wind disturbance in hover mode. It can be observed that when the amplitude of the head-on wind disturbance is around 5 m/s, the aircraft with the SC can finally stabilize in place, whereas the aircraft with only the VRC is continuously blown toward the negative direction of the x_g -axis in the ground frame by external wind disturbance force. Although the SC causes larger oscillations in the aircraft's y_g -axis displacement under the disturbance of the randomly fluctuating $v_{gust,r}^g$ compared with the VRC, similar to the case of Euler angles, the amplitude of the undesirable y_g -axis displacement shown in Fig. 9(n) is <0.1 m, which is considered an acceptable adverse effect.

Fig. 10(m) to Fig. 10(o) present the displacement feedback data of the aircraft under local wind disturbance in hover mode. It can be seen that under this test condition, the adoption of the SC enables the aircraft to restore its original position more rapidly along the x_g -axis. Although the recovery along the y_g -axis is slightly slower, the overall variation trend is smoother, while the oscillation is more pronounced when the VRC is adopted.

Fig. 11(m) to Fig. 11(o) present the displacement data of the aircraft in the global coordinate system from the transition phase to the cruise mode, where a positive x-axis displacement indicates forward flight of the aircraft, a positive y-axis displacement indicates rightward flight, and a negative z-axis displacement indicates upward flight. The reason for the y-axis displacement is the persistent rightward lateral wind disturbance. This study focuses on disturbance rejection control during the hover mode and transition phase, and does not involve trajectory tracking control in the cruise mode for the time being; thus, closed-loop control is not applied to the displacement data in the cruise mode.

Analysis of limit disturbance rejection performance:

In Fig. 9(p), multiple simulations were performed by varying the amplitude K_c of the head-on wind disturbance at a step size of 0.1, and the cumulative data of the x_g -axis error throughout the full simulation duration were recorded. When the aircraft can stabilize near the hovering point, the slope of the cumulative error data with respect to wind speed is low; once the aircraft fails to maintain its original position, the slope abruptly increases to a larger value. The curve inflection point (CIP) where the slope changes abruptly can be used to measure the limit head-on wind disturbance rejection performance of the aircraft. For the VRC alone, the inflection point is located at $K_c=4.2$, while for the SC, the inflection point shifts to $K_c=6.3$. Since the dynamic pressure of wind disturbance is proportional to the square of its velocity, the SC method can improve the head-on wind disturbance rejection performance of the aircraft by approximately 125%.

In Fig. 10(p), multiple simulations are performed by varying the amplitude K_c of the left wing, but the cumulative error is calculated using the axis angle error in Eq. (83) instead, so as to intuitively demonstrate the level of local wind disturbance under which the aircraft will experience violent loss of control. The CIP is at $K_c=4.8$ for the VRC, while it shifts to $K_c=7.8$ for the SC. Therefore, the SC can improve the local gust rejection performance of the aircraft by approximately 164%.

Similarly, it can be seen from Fig. 11(p) that the CIP is located at $K_c=4.4$ for the VRC, while it shifts to $K_c=6.1$ for the SC. Therefore, the SC can improve the local gust rejection performance of the aircraft during the transition phase by approximately 92.2%.

5.1.3. Parameter variation analysis

To more accurately characterize the performance improvement of the proposed configuration and control method for tail-sitter UAVs during the hover mode and transition phase, the number of samples with reproducible random number seeds is increased to 10 in this subsection, and the aforementioned three sets of limit disturbance rejection performance analysis experiments are repeated. For all simulation results, curves are plotted and CIP positions are recorded in accordance with the analysis approach applied in Fig. 9(p), Fig. 10(p) and Fig. 11(p). The improvement percentage (abbreviated as Imp. % in the table) of all SC cases relative to the VRC cases is calculated, and the corresponding

average value (abbreviated as Avg. in the table) is obtained via the extremum-removal averaging method, with the summarized data presented in Table 3.

In summary, the proposed SC method exhibits an outstanding improvement in disturbance rejection performance for tail-sitter UAVs during both hover mode and transition phase. Compared with the VRC method, the SC achieves a theoretical improvement of approximately 114.7% in head-on wind disturbance rejection performance and approximately 171.2% in local wind disturbance rejection performance for the aircraft in hover mode, along with a theoretical improvement of approximately 105.1% in disturbance rejection performance during transition phase.

Notably, the theoretical results of this statistical analysis are only valid for the 1-cosine discrete gust model established in Eq. (91) to Eq. (93), excluding scenarios with extreme wind fields featuring severe wind speed and direction fluctuations, as this class of tail-sitter UAVs is inherently not designed for flight operations in extreme weather. In addition, all theoretical data are derived from the configuration parameters of the PW-TUAV proposed herein, which are only referenceable for homogeneous tail-sitter UAVs with a wingspan of approximately 1 m and a total weight of around 0.5 kg, and cannot be directly extended to tail-sitter UAV platforms with heavy payloads, high aspect ratios, or other specialized configurations.

5.2. Flight validation of PW-TUAV prototype

5.2.1. Feedback control system deployment on prototype

The first-generation prototype of the PW-TUAV has been fabricated, and its structural and material properties can be referred to in Fig. 1. The onboard controller is designed based on the STM32F405RGT6 micro-controller chip, with the main control frequency set to 100 Hz. It is connected to four components via four serial port communications, respectively: the WitMotion JY901B IMU sensor (integrated with the MPU-9250 chip and BMP280 barometer), the FlySky i6x-a8s receiver adopting the SBUS protocol, the Micoair 2.4 G LoRa module, and the MTF-01 optical flow ranging integrated module. Among them, the data transmission frequency of the LoRa module to the ground station is 20 Hz, while the frequencies of all other communications are 100 Hz. In addition, the WitMotion JY-DATA-LOG-USB data recorder is mounted onboard, capable of storing flight logs at a maximum rate of 90 kb/s, which shares one serial port with the LoRa module. Fig. 12 presents the deployment flow block diagram of the complete feedback control system implemented on the PW-TUAV prototype, along with detailed state variables transferred between individual modules. Programmed with the DSP library, the system achieves a single-loop execution time of <100 μ s, which is sufficient to satisfy the demands of high-frequency control.

The following is a detailed introduction to the control flow shown in Fig. 12.

The desired pose signals are generated in two ways, determined by the flight mode of the aircraft. When the aircraft is in VTOL or hover mode, the desired position signal and yaw angle signal are obtained from remote controller commands, and the desired pitch angle and roll angle signals are acquired from the position control loop shown in Fig. 8. After switching the flight mode stick on the remote controller, the aircraft automatically enters the transition phase, at which point the transition trajectory planner is called to generate the desired attitude signals. Since this study focuses on verifying the disturbance rejection performance of the aircraft in hover mode and transition phase, rather than the optimization algorithm of transition trajectories, a simple transition trajectory planner that only concerns the aircraft attitude and is easy to implement is designed. The transition trajectory planner runs only once at the instant when the flight mode switching command is received. Based on the deviation between the current attitude and the target attitude, it automatically calculates the transition duration with a maximum angular rate limit of $20^\circ/\text{s}$, and generates a linear, smooth

Table 3
Statistical analysis of the ultimate disturbance rejection performance of aircraft.

Sample number	1	2	3	4	5	6	7	8	9	10	Avg.
Random seed	123	189	255	321	387	453	519	585	651	717	
Exp. 1: CIP of VRC	4.2	4.2	4.2	4.2	4.5	4.3	4.3	4.0	4.2	4.2	
Exp. 1: CIP of SC	6.3	6.2	6.2	6.1	6.4	6.1	6.2	6.1	6.1	6.3	
Exp. 1: Imp. %	125.0	117.9	117.9	110.9	102.3	101.2	107.9	132.6	110.9	125.0	114.7
Exp. 2: CIP of VRC	4.8	4.7	4.7	4.8	4.9	4.7	4.8	4.5	4.7	4.7	
Exp. 2: CIP of SC	7.8	7.8	7.8	7.8	7.8	7.8	7.8	7.8	7.8	7.8	
Exp. 2: Imp. %	164.0	175.4	175.4	164.0	153.4	175.4	164.1	200.4	175.4	175.4	171.2
Exp. 3: CIP of VRC	4.4	4.1	4.4	4.3	4.3	4.3	4.2	4.3	4.4	4.1	
Exp. 3: CIP of SC	6.1	6.2	6.2	6.2	6.0	6.1	6.1	6.2	6.2	6.1	
Exp. 3: Imp. %	92.2	128.7	98.6	107.9	94.7	101.2	110.9	107.9	98.6	121.4	105.1

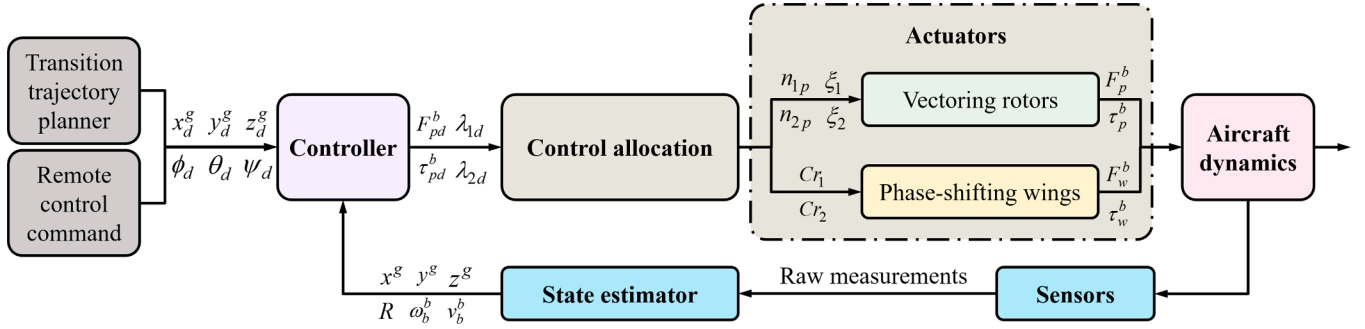


Fig. 12. Flow block diagram of the feedback control system for the PW-TUAV prototype.

commanded attitude transition trajectory with continuous output and no abrupt jumps. The planned trajectory is called in sequence by the control loop until the target attitude is reached. After the transition phase ends, the aircraft enters cruise mode, and the desired signal generation method switches back to real-time remote controller commands.

Details of the Controller module have been presented in Fig. 8, which processes the desired and feedback pose data and outputs the desired forces and moments for the vectoring rotor, as well as the phase angles of the wings. The Control allocation module converts the upper-level desired commands into actual controller command signals according to the inverse dynamic mapping relationships shown in Fig. 4 and Fig. 6, namely the motor speed n_{ip} and rotor tilt angle ξ_i of the vectoring rotor mechanism, and the crank angle Cr_i of the phase-shifting wing mechanism. These command signals are converted into PWM signals to drive each actuator, generating the corresponding control forces and moments.

The sensors mainly include an IMU module and an OFS. The IMU module is integrated with the micro flight control board and fixed at the front of the airframe, while the OFS is separately mounted at the bottom of the wing along the z_b -axis. The OFS equipped on the PW-TUAV prototype integrates an optical flow camera and a ranging module, which can provide 2D pixel motion data and altitude measurement data simultaneously.

The state estimator is used to process the raw data from various sensors. The accelerometer, gyroscope and magnetometer data of the IMU are processed via the Kalman filter algorithm to obtain angular velocity and Euler angle data, which form the real-time attitude matrix through DCM conversion. Subsequently, based on the Euler angle and angular velocity information respectively, the rotational parasitic errors of the OFS measurements in vertical altitude and horizontal velocity are eliminated, with only the effective translational terms retained. The effective horizontal velocity is integrated to obtain horizontal displacement data relative to the ground. The final output of the state estimator is determined by the aircraft's current flight mode. When the aircraft is in takeoff, landing or hover mode, the state variables fed back to the controller include ground displacement and velocity, as well as

the attitude matrix and angular velocity. When the aircraft is in the transition phase or cruise mode, since the present study does not cover trajectory optimization algorithms and cruise position control, the only state variables fed back to the controller at this time are the attitude matrix and angular velocity. This corresponds to the invocation mode of the desired signals.

5.2.2. Outdoor flight experiment demonstration

Fig. 13 presents the experimental results of the PW-TUAV prototype during long-duration outdoor flight in hover mode, while Fig. 14 records the complete process from hover mode, through the transition phase, to cruise mode. All flight experiments were conducted under the disturbance of approximately level 3 natural wind, with a potential local peak wind speed of around 5.4 m/s.

The hover experiment lasted for approximately half a minute. After takeoff, the PW-TUAV prototype was set to maintain fixed-point and fixed-altitude flight at an altitude of around 1 m, and Fig. 13(a) presents the time series of the pose at an interval of approximately 5 s. The experimental data demonstrate that within the first 20 s, the three-axis Euler angles of the PW-TUAV prototype were stabilized within 2.8° , 13.7° and 13.2° respectively, with the maximum lateral displacement velocity not exceeding 0.24 m/s. Around the 24 s mark, a significant disturbance from local lateral strong wind occurred, which induced an instantaneous fluctuation in the yaw angle of the PW-TUAV prototype with a peak value of approximately 30° . This event was accompanied by a roughly 10° fluctuation in the roll angle and a roughly 20° fluctuation in the pitch angle, which corresponded to the desired attitudes generated to maintain lateral position stability. Even in this case, the maximum instantaneous lateral displacement velocity did not exceed 0.5 m/s. It can also be seen from the pose time series that both the altitude and the lateral displacement of the aircraft are extremely small. Furthermore, the PW-TUAV prototype restored attitude stability within approximately 3 s, exhibiting excellent stability in hover mode.

Fig. 14(a) to Fig. 14(d) present the pose sequences at key time nodes of the PW-TUAV prototype, from vertical takeoff mode, to entering the transition phase, through the end of the transition phase, and finally to entering cruise mode. The aircraft took off from the ground at time 0,

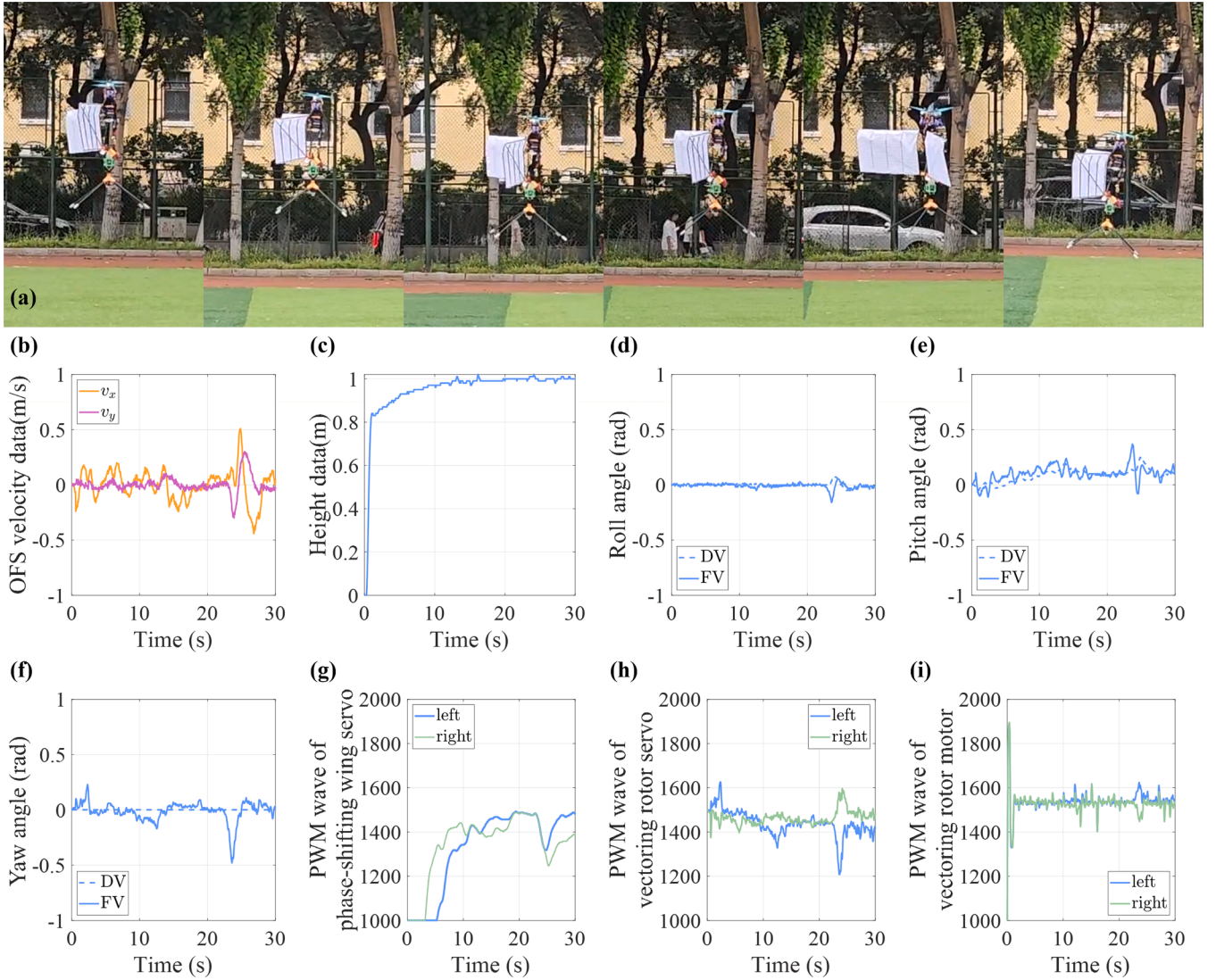


Fig. 13. Hover flight experimental results of the PW-TUAV prototype. (a) Time series of the aircraft's hover pose at an interval of approximately 5 s. Since the sequential images were manually extracted, there may be a slight temporal deviation. (b) OFS horizontal velocity data. (c) Altitude data. (d)–(f) Euler angle data. (g)–(i) PWM-form control signals for each actuator.

with a continuously increasing desired altitude setpoint. The transition flight command was received at approximately 3.7 s, and after the pitch angle was gradually adjusted to -70° over a period of around 3.3 s, the aircraft entered the cruise flight state. The experimental data demonstrate that even when the PW-TUAV prototype was subjected to high-altitude airflow disturbance during the transition phase, which induced a multi-axis coupled disturbance of up to 15.5° in both the roll angle and yaw angle, it could quickly return to a stable state. This verifies the safe and stable takeoff capability of the proposed design under the disturbance of level 3 natural wind, which is challenging for this type of tail-sitter UAVs with a wingspan of 1 m but a self-weight of <0.5 kg.

5.3. Limitations and future work

In this section, we discuss the existing limitations of this study from several aspects and briefly outline the future work plan.

First, in terms of configuration design: the current PW-TUAV prototype mainly adopts a simple flat airfoil or a single-sided curved airfoil with a slight camber. This is mainly to meet the lightweight design requirements: the wing is composed solely of hollow carbon fiber rods and P31N parachute fabric, with a mass of only approximately 17 g for a single wing, accounting for merely around 3.5% of the prototype's total

self-weight. In this study, as the core focus is to improve the disturbance rejection performance of the aircraft in hover mode and transition phase, the use of this type of wing is provisionally appropriate. However, in cruise mode, this type of airfoil may result in an extremely narrow cruise flight envelope, which only enables low-speed level flight and cannot fulfill long-endurance cruise flight missions with a wide speed range. In the future, we plan to design the wing with an inner-outer two-segment structure: the inner segment adopts a rigid airfoil structure similar to that of fixed-wing aircraft, while the outer segment continues to use the bionic phase-shifting wings proposed in this study. This improvement is expected to achieve a balanced trade-off between the flight efficiency and disturbance rejection performance of the aircraft, as the inner-segment airfoil will provide superior lift-drag characteristics, while the outer-segment airfoil can still deliver a high auxiliary control moment due to its larger spanwise moment arm.

Second, in terms of dynamic modeling: the finite element-based QS reduced-order model adopted in this study is mainly used to address the distribution of aerodynamic circulation forces on the wing under complex rotational maneuvers. To reduce the required computational resources, complex aerodynamic phenomena, including rotational circulation, added mass, aerodynamic interference effects from rotor slipstream, and aeroelastic effects, are neglected in the model.

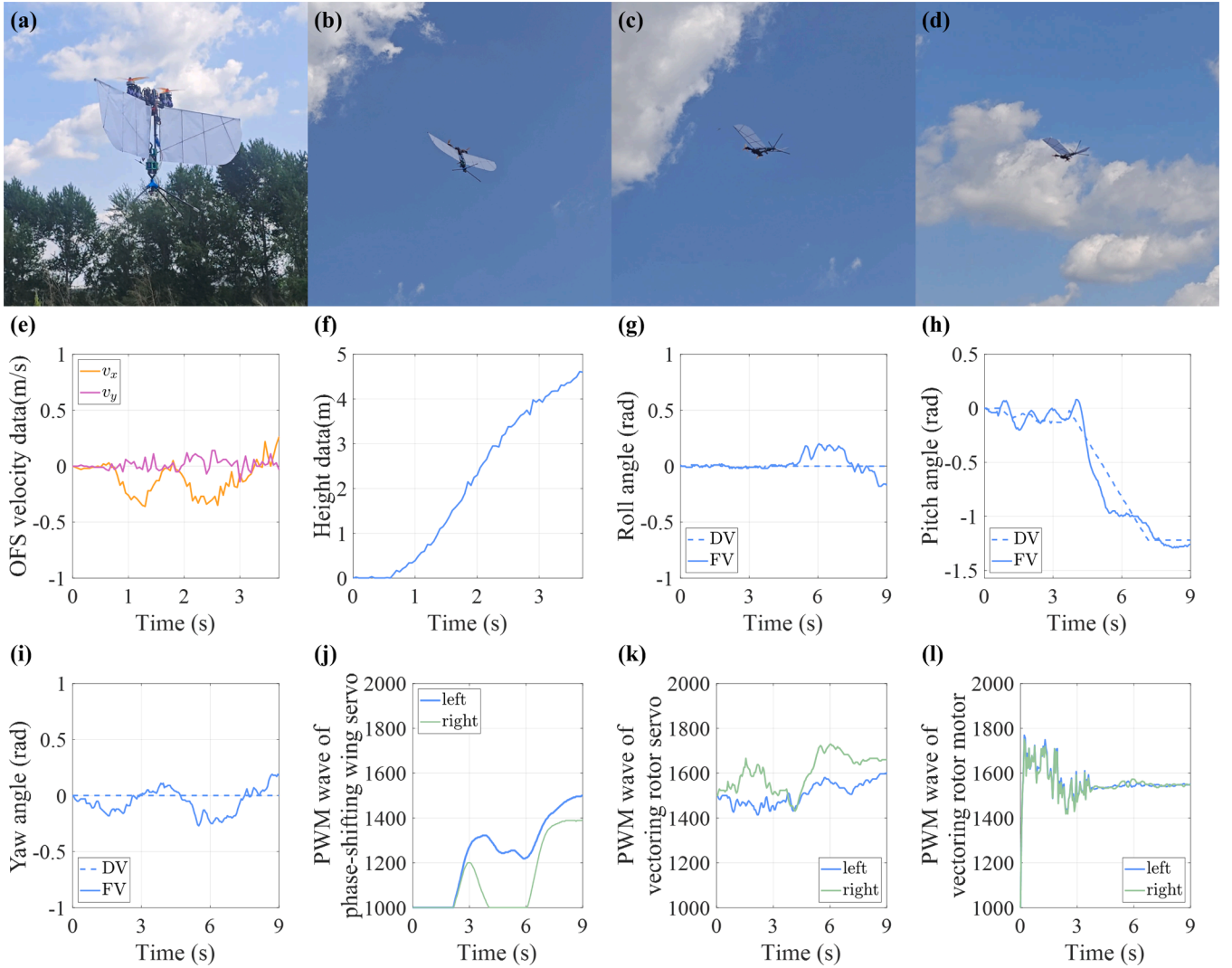


Fig. 14. Transition flight experimental results of the PW-TUAV prototype. (a) The aircraft undergoing vertical takeoff. (b) The aircraft entering the transition phase. (c) The aircraft completing the transition. (d) The aircraft performing cruise mode. (e) OFS horizontal velocity data during the vertical takeoff phase. (f) Altitude data during the vertical takeoff phase. (g)–(i) Euler angle data of the aircraft throughout the process from vertical takeoff to transition phase. (j)–(l) PWM wave-form control signals of each actuator.

Accordingly, there exists a certain deviation between the actual aerodynamic loads and the values predicted by the model. However, pursuing higher accuracy of the aerodynamic model will inevitably lead to an increase in computational cost, which is an intractable bottleneck problem in the current field of aerodynamic modeling technology. For future work, we plan to conduct high-precision wind tunnel tests to measure the aerodynamic characteristics of the prototype, and further correct the aerodynamic model using empirical coefficients fitted from the experimental data.

Third, in terms of controller design: this study proposes an easy-to-implement SC design scheme, which is highly conducive to the preliminary research and exploration of this type of aircraft. It adopts a simple integral-driven dynamic weight allocation strategy, as only the most basic IMU and OFS are equipped as airborne sensors. The technical limitations of the difficult miniaturization of existing airborne sensors make it extremely challenging to deploy pressure sensors on the wings of small UAVs. Although the integral-driven dynamic weight allocation strategy presented in this study has been verified to deliver a certain improvement in disturbance rejection performance, the performance potential of the phase-shifting wing mechanism has in fact not yet been efficiently exploited to a fuller extent due to the constrained sensing capability. For future work, we plan to design a dedicated wind

disturbance observer to optimize the flight control system, so as to achieve rapid feedforward compensation for high-frequency, sudden gust disturbances and further optimize the control performance.

Finally, in terms of flight experiments: this study mainly presents experimental results in hover mode and transition phase, as the current version of the PW-TUAV prototype still has several unresolved issues in the cruise mode, such as the inability to adjust flight speed and difficulty in altitude control. This is because the control torque of the vectoring rotor mechanism is highly dependent on propeller thrust; once the propeller thrust decreases, the aircraft is prone to loss of control due to its non-minimum-phase characteristics. Therefore, the experimental performance in the cruise mode is not yet satisfactory. We clarify that the first-generation prototype of the PW-TUAV is not yet a fully mature design solution, but it provides a novel and effective approach to enhance the disturbance rejection performance of tail-sitter UAVs. In the future, we will strive to further improve the cruise mode performance of the PW-TUAV prototype. Increasing the control arm length may be the simplest solution to address the insufficient disturbance rejection performance in the cruise mode, but we consider this an inefficient measure that is also not conducive to the structural compactness design indicator. We are attempting to develop the complex bionic deformation capability of the wings to optimize attitude control performance more efficiently.

Meanwhile, the phase-shifting wing mechanism proposed in this study also has further development potential in dynamically adjusting the dihedral angle during the cruise mode to assist in the aircraft's maneuvering and turning control. Additional sensors including GPS and airspeed indicators will be integrated to conduct experimental tests on the maneuvering and turning performance of the PW-TUAV prototype in cruise mode.

6. Conclusion

Focusing on the disturbance rejection control challenges faced by tail-sitter UAVs during the hover mode and transition phase, this study proposes a novel PW-TUAV configuration, and completes a series of work including configuration design, dynamic modeling, response characteristic analysis, SC development, and experimental validation. By means of a simple yet effective linkage-driven mechanism, this configuration addresses the contradiction between the large aerodynamic load torque acting on the wing root and the insufficient driving torque of micro servos, thereby enabling the aircraft to achieve a balanced trade-off between lightweight design requirements and structural stability. The established multibody dynamics model fully accounts for the time-varying characteristics of the wing aerodynamic center, unsteady aerodynamic forces induced by rotational motion, and the non-minimum-phase characteristics of vectoring rotors; in comparison with single-rigid-body models, it can more accurately reflect the system dynamics under complex flight conditions. A full-attitude controller is designed based on the SO(3) space to support large-angle pitch-axis maneuvering of the aircraft during the transition phase, and an integral-driven weight allocation strategy is proposed to realize synergistic control of the vectoring rotor mechanisms and phase-shifting wing mechanisms. Simulation results verify that the proposed SC achieves a significant performance improvement in the disturbance rejection capability of this class of aircraft with an approximate 1 m wingspan

and 0.5 kg total weight during hover mode and transition phase, compared with the scheme adopting only the VRC. The fabricated initial PW-TUAV prototype successfully completes the entire flight process from VTOL to transition to cruise mode in outdoor experiments, demonstrating safe and stable take-off capability under level-3 natural wind disturbance conditions. Overall, this study provides a novel and effective approach to enhance the disturbance rejection performance of tail-sitter UAVs during hover mode and transition phase.

CRedit authorship contribution statement

Tianyou Mao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bosong Duan:** Methodology, Funding acquisition. **Feiyu Liu:** Validation, Investigation. **Chuangqiang Guo:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Abbreviations

The following abbreviations are used in this manuscript:

Abbreviations	Full name
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Take-Off and Landing
PW-TUAV	Phase-shifting Wing Tail-sitter Unmanned Aerial Vehicle
SO(3)	Special Orthogonal Group in three dimensions
QS	Quasi-Steady
AR	Aspect Ratio
DCM	Direction Cosine Matrix
PSO	Particle Swarm Optimization
IMU	Inertial Measurement Unit
OFS	Optical Flow Sensor
DV	Desired Value
FV	Feedback Value
SC	Synergistic Controller
VRC	Vectoring Rotor Controller
CIP	Curve Inflection Point

Appendix B. The notation for kinematic variables

Throughout the entire derivation process of the aircraft kinematics and dynamics: m denotes mass, J denotes moment of inertia, F denotes force, τ denotes torque, and r denotes position vector. ω denotes angular velocity, v denotes linear velocity, $\dot{\omega}$ denotes angular acceleration, and $\dot{v}$ denotes linear acceleration. Superscripts indicate the coordinate frame in which the physical quantity is expressed: b represents the body frame, w_i represents the wing frame, and pi represents the vectoring rotor frame, where $i = 1$ denotes the left side and $i = 2$ denotes the right side. Subscripts are defined in relation to the physical quantities themselves. Subscripts of m and J indicate the corresponding component. For F and τ , a single-letter subscript denotes the resultant force or resultant torque acting on the component; if the subscript consists of two letters separated by a comma, the first letter indicates the type of force or torque, and the second letter indicates the component on which the force or torque acts. The definition of the

characteristic velocity U is consistent with that of force and torque. For ω , a single-letter subscript denotes the resultant angular velocity of the component; if the subscript consists of two letters separated by a comma, it denotes the relative angular velocity of the coordinate frame indicated by the first letter with respect to that indicated by the second letter. The subscript definitions for linear velocity, angular acceleration, and linear acceleration are consistent with that for angular velocity. Subscripts of r generally consist of two letters separated by a comma, representing the position vector pointing from the entity indicated by the first letter to that indicated by the second letter.

Appendix C. Four-bar linkage vector loop equation

The vector loop equation can be expressed as:

$$L_1 e^{i\theta_1} - L_2 e^{i\theta_2} - L_3 e^{i\theta_3} + L_4 e^{i\theta_4} = 0 \quad (94)$$

It can be simplified to:

$$A_1 \cos \theta_4 + A_2 \sin \theta_4 + A_3 = 0 \quad (95)$$

Where:

$$\begin{cases} A_1 = 2L_4(L_1 \cos \theta_1 - L_2 \cos \theta_2) \\ A_2 = 2L_4(L_1 \sin \theta_1 - L_2 \sin \theta_2) \\ A_3 = L_1^2 + L_2^2 + L_4^2 - L_3^2 - 2L_1 L_2 \cos(\theta_1 - \theta_2) \end{cases} \quad (96)$$

Solving this equation yields:

$$\theta_4 = \text{atan2}(A_2, A_1) \pm \arccos\left(-\frac{A_3}{\sqrt{A_1^2 + A_2^2}}\right) \quad (97)$$

Substituting θ_4 back into Eq. (94) gives the solution:

$$\theta_3 = \arg(L_1 e^{i\theta_1} - L_2 e^{i\theta_2} + L_4 e^{i\theta_4}) \quad (98)$$

The angle of the wing is:

$$\theta_5 = \theta_4 + \text{atan}\left(\frac{L_6}{L_5}\right) \quad (99)$$

The first-order derivative of the vector loop equation with respect to time can be expressed as:

$$\begin{bmatrix} -L_3 \sin \theta_3 & L_4 \sin \theta_4 \\ L_3 \cos \theta_3 & -L_4 \cos \theta_4 \end{bmatrix} \begin{bmatrix} \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix} = \dot{\theta}_2 \begin{bmatrix} L_2 \sin \theta_2 \\ -L_2 \cos \theta_2 \end{bmatrix} \quad (100)$$

It can be solved as:

$$\begin{cases} \dot{\theta}_3 = \frac{L_2 \sin(\theta_2 - \theta_4)}{L_3 \sin(\theta_3 - \theta_4)} \dot{\theta}_2 \\ \dot{\theta}_4 = \frac{L_2 \sin(\theta_2 - \theta_3)}{L_4 \sin(\theta_4 - \theta_3)} \dot{\theta}_2 \end{cases} \quad (101)$$

The second-order derivative of the vector loop equation with respect to time can be expressed as:

$$\begin{bmatrix} -L_3 \sin \theta_3 & L_4 \sin \theta_4 \\ L_3 \cos \theta_3 & -L_4 \cos \theta_4 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_3 \\ \ddot{\theta}_4 \end{bmatrix} = \begin{bmatrix} ACC_X \\ ACC_Y \end{bmatrix} \quad (102)$$

$$\begin{cases} ACC_X = L_2 \ddot{\theta}_2 \sin \theta_2 + \sum_{i=2}^4 c_i L_i \dot{\theta}_i^2 \cos \theta_i \\ ACC_Y = -L_2 \ddot{\theta}_2 \cos \theta_2 + \sum_{i=2}^4 c_i L_i \dot{\theta}_i^2 \sin \theta_i \end{cases} \quad (103)$$

where $c_2 = 1$, $c_3 = 1$, $c_4 = -1$. The roots of the equations can be obtained as follows:

$$\begin{cases} \ddot{\theta}_3 = \frac{-L_4 \cos \theta_4 ACC_X - L_4 \sin \theta_4 ACC_Y}{L_3 L_4 \sin(\theta_3 - \theta_4)} \\ \ddot{\theta}_4 = \frac{-L_3 \cos \theta_3 ACC_X - L_3 \sin \theta_3 ACC_Y}{L_3 L_4 \sin(\theta_3 - \theta_4)} \end{cases} \quad (104)$$

Appendix D. Control parameters tuning

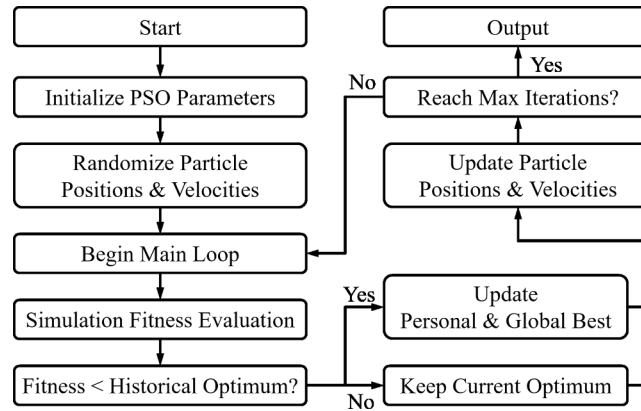


Fig. 15. PSO algorithm flowchart.

The implementation flow of the PSO algorithm is shown in Fig. 15. Specifically, each particle is defined as a 9-dimensional reference parameter vector, which corresponds to the proportional, integral and derivative coefficients of the three rotation axes respectively. The population size is set to 50, the maximum iteration number of the algorithm is set to 50, the inertia weight is set to 0.7, and both the individual cognitive coefficient and social learning coefficient are set to 1.5. For the reference parameters of each particle in the population, uniformly distributed random perturbations are added within the range of -100% to $+500\%$ to generate the initial particle positions.

In addition, the controller proposed in this study is designed based on the Lie group and Lie algebra framework, and its attitude tracking error is uniformly quantified using the globally singularity-free axis-angle error. This is slightly different from the traditional Euler angle-based channel-by-channel evaluation mode. Accordingly, a multi-objective weighted fitness function tailored to this error description system is constructed. The performance evaluation indicators are designed as follows:

(1) Steady-state performance indicator: Based on the axis-angle error in the steady-state interval of the simulation, it quantifies the steady-state control accuracy and smoothness of the system; (2) Overshoot indicator: Based on the peak value of the axis-angle error, the overshoot ratio is calculated to constrain the maximum overshoot amplitude of the attitude response; (3) Global tracking accuracy indicator: Based on the integral of absolute error of the axis-angle error over the full time domain, it evaluates the full-process tracking accuracy of the system; (4) Settling time indicator: Based on the time taken for the error to enter the steady-state tolerance band, it characterizes the steady-state convergence speed of the system; (5) Rise time indicator: Based on the time taken for the error to drop to the specified threshold, it measures the initial response speed of the system; (6) Response smoothness indicator: Based on the non-monotonic fluctuation degree of the error during the adjustment process, it evaluates the smoothness of the attitude response; (7) Abnormality penalty term: A heavy penalty is imposed on working conditions such as simulation divergence and system non-convergence to eliminate engineering-infeasible parameters.

The above multi-dimensional performance indicators are integrated into a minimization objective function via the linear weighting method. In the main iteration loop, the fitness values of all particles are calculated, the individual and global optimal solutions are compared and updated, and the iterative update of the particle positions and velocities is completed. The algorithm terminates when the number of iterations reaches the preset maximum value, and the globally optimal control parameter set is output.

Data availability

Data will be made available on request.

References

- [1] J. Liscouët, F. Pollet, J. Jézégou, M. Budinger, S. Delbecq, J.-M. Moschetta, A methodology to integrate reliability into the conceptual design of safety-critical multirotor unmanned aerial vehicles, *Aerosp. Sci. Technol.* 127 (2022) 107681, <https://doi.org/10.1016/j.ast.2022.107681>.
- [2] B. Yang, J. Lu, T. Liu, B. Zhang, C. Geng, Y. Tian, S. Zhang, Traffic accident rescue action recognition method based on real-time UAV video, *Drones* 9 (8) (2025) 519, <https://doi.org/10.3390/drones9080519>.
- [3] C. Chen, S. Chen, G. Hu, B. Chen, P. Chen, K. Su, An auto-landing strategy based on pan-tilt based visual servoing for unmanned aerial vehicle in GNSS-denied environments, *Aerosp. Sci. Technol.* 116 (2021) 106891, <https://doi.org/10.1016/j.ast.2021.106891>.
- [4] Y. Liu, Z. Ge, Y. Yuan, X. Su, X. Guo, T. Suo, Q. Yu, Wing deformation measurement using the stereo-vision methods in the presence of camera movements, *Aerosp. Sci. Technol.* 119 (2021) 107161, <https://doi.org/10.1016/j.ast.2021.107161>.
- [5] A. Divazi, R. Askari, E. Roohi, Experimental and numerical investigation on the spraying performance of an agricultural unmanned aerial vehicle, *Aerosp. Sci. Technol.* 160 (2025) 110083, <https://doi.org/10.1016/j.ast.2025.110083>.
- [6] Y. Liu, H. Feng, Y. Fan, J. Yue, F. Yang, J. Fan, Y. Ma, R. Chen, M. Bian, G. Yang, Utilizing UAV-based hyperspectral remote sensing combined with various agronomic traits to monitor potato growth and estimate yield, *Comput. Electron. Agric.* 231 (2025) 109984, <https://doi.org/10.1016/j.compag.2025.109984>.
- [7] W. Yue, X. Zhang, Hawk-rabbit game architecture for unmanned aerial vehicle swarm multi-target defense under uncertain attack targets, *Aerosp. Sci. Technol.* 164 (2025) 110379, <https://doi.org/10.1016/j.ast.2025.110379>.
- [8] Y. Fu, B. Wang, H. Zhao, M. Zhou, N. Li, Z. Gao, Adaptive safety attitude control of a hybrid VTOL UAV under transition flight subject to multiple faults and uncertainties, *Aerosp. Sci. Technol.* 163 (2025) 110284, <https://doi.org/10.1016/j.ast.2025.110284>.
- [9] B. Duan, C. Guo, T. Mao, H. Liu, Wing geometry and kinematic parameters optimization of bat-like robot fixed-altitude flight for minimum energy, *Aerosp. Sci. Technol.* 140 (2023) 108482, <https://doi.org/10.1016/j.ast.2023.108482>.
- [10] Y. Shen, Z. Wang, K. Wang, Z. Zhou, X. Zou, Propulsion/aerodynamic coupling analytic modeling of distributed propulsion wing unit with induced wing, *Aerosp. Sci. Technol.* 168 (2026) 110883, <https://doi.org/10.1016/j.ast.2025.110883>.
- [11] B. Crowther, E. Koning, V. Crann, C. Marsh, D. Usov, K. Wood, K. Kabbabe, A. Filippone, I. Lunnon, Analytical and experimental evaluation of wind speed limits for a quadplane UAS in hover, *AIAA SCITECH 2025 Forum, American Institute of Aeronautics and Astronautics, Orlando, FL, 2025*, <https://doi.org/10.2514/6.2025-1827>.
- [12] A. Mathur, E. Atkins, Energy aware coverage planning for a quadplane small uncrewed aircraft system, in: 2025 International Conference on Unmanned Aircraft Systems (ICUAS), Charlotte, NC, USA, IEEE, 2025, pp. 929–936, <https://doi.org/10.1109/ICUAS65942.2025.11007898>.
- [13] A. Mathur, E. Atkins, Multi-mode flight simulation and energy-aware coverage path planning for a lift+cruise quadplane, *Drones* 9 (4) (2025) 287, <https://doi.org/10.3390/drones9040287>.

- [14] M. Okulski, M. Ławryńczuk, A small UAV optimized for efficient long-range and VTOL missions: an experimental tandem-wing quadplane drone, *Appl. Sci.* 12 (14) (2022) 7059, <https://doi.org/10.3390/app12147059>.
- [15] M. Allenspach, G.J.J. Ducard, Nonlinear model predictive control and guidance for a propeller-tilting hybrid unmanned air vehicle, *Automatica* 132 (2021) 109790, <https://doi.org/10.1016/j.automatica.2021.109790>.
- [16] Z. Liang, L. Fan, G. Wen, Z. Xu, Design, modeling, and control of a composite tilt-rotor unmanned aerial vehicle, *Drones* 8 (3) (2024) 102, <https://doi.org/10.3390/drones8030102>.
- [17] E.U. Rojo-Rodriguez, E.G. Rojo-Rodriguez, S.A. Araujo-Estrada, O. Garcia-Salazar, Design and performance of a novel tapered wing tiltrotor UAV for hover and cruise missions, *Machines* 12 (9) (2024) 653, <https://doi.org/10.3390/machines12090653>.
- [18] K. Xiao, Y. Meng, X. Dai, H. Zhang, Q. Quan, A lifting wing fixed on multirotor UAVs for long flight ranges, in: 2021 International Conference on Unmanned Aircraft Systems (ICUAS), Athens, Greece, IEEE, 2021, pp. 1605–1610, <https://doi.org/10.1109/ICUAS51884.2021.9476859>.
- [19] H. Wei, S. Wang, Q. Quan, Tethered lifting-wing multicopter landing like kite, in: 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, IEEE, 2024, pp. 11811–11817, <https://doi.org/10.1109/ICRA57147.2024.10611536>.
- [20] X. Zhao, Z. Zhou, J. Hu, K. Wang, B. Li, Dynamics analysis, control and flight test of all-wing tail-sitter configuration solar powered UAV, *Aerosp. Sci. Technol.* 168 (2026) 110945, <https://doi.org/10.1016/j.ast.2025.110945>.
- [21] D. Wang, J. Xu, Z. Qiao, Y. Bai, C. Peng, Cascaded finite-time hovering attitude control for a biplane quadrotor tail-sitter UAV considering actuator dynamics, *Aerosp. Sci. Technol.* 168 (2026) 110780, <https://doi.org/10.1016/j.ast.2025.110780>.
- [22] G. Lu, Y. Ren, F. Zhu, H. Li, R. Xue, Y. Cai, X. Lyu, F. Zhang, Autonomous tail-sitter flights in unknown environments, *IEEE Trans. Robot.* 41 (2025) 1098–1117, <https://doi.org/10.1109/TRO.2025.3526102>.
- [23] O. Juhasz, J.-P. Redding, System identification of a hovering quadrotor biplane tailsitter with canted motors, in: Proceedings of the Vertical Flight Society 80th Annual Forum, Montreal, Canada, The Vertical Flight Society, 2024, pp. 1–12, <https://doi.org/10.4050/F-0080-2024-1190>.
- [24] G.J.J. Ducard, M. Allenspach, Review of designs and flight control techniques of hybrid and convertible VTOL UAVs, *Aerosp. Sci. Technol.* 118 (2021) 107035, <https://doi.org/10.1016/j.ast.2021.107035>.
- [25] K. Murali, E.P. Moreno, L.R. Lustosa, Flight validation of a global singularity-free aerodynamic model for flight control of tail sitters, in: 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, IEEE, 2024, pp. 3861–3867, <https://doi.org/10.1109/ICRA57147.2024.10610780>.
- [26] E. Tal, G. Ryou, S. Karaman, Aerobatic trajectory generation for a VTOL fixed-wing aircraft using differential flatness, *IEEE Trans. Robot.* 39 (6) (2023) 4805–4819, <https://doi.org/10.1109/TRO.2023.3301312>.
- [27] Z. Ma, E.J.J. Smeur, G.C.H.E. De Croon, Design and control of a tilt-rotor tailsitter aircraft with pivoting VTOL capability, *IEEE Robot. Autom. Lett.* 10 (6) (2025) 5911–5918, <https://doi.org/10.1109/LRA.2025.3563821>.
- [28] W. Di, Z. Wei, Q. Liu, J. Zhang, J. Xiang, D. Li, Z. Tu, Dynamics and control of tailless vectored-thrust VTOL in hover-to-forward transition, *Aerosp. Sci. Technol.* 164 (2025) 110385, <https://doi.org/10.1016/j.ast.2025.110385>.
- [29] R. D'Sa, N. Papanikolopoulos, Design and experiments for multi-section-transformable (MIST)-UAV, in: 2019 International Conference on Robotics and Automation (ICRA), Montreal, QC, Canada, IEEE, 2019, pp. 1878–1883, <https://doi.org/10.1109/ICRA.2019.8793575>.
- [30] Q.C. Chen, Z.W. Shi, W.Y. Zhang, et al., Aerodynamic layout and control strategy design and flight verification of a wing-foldable variant VTOL aircraft, *Acta Astronaut. Astronaut. Sinica* 45 (6) (2024) 629583, <https://doi.org/10.7527/S1000-6893.2023.29583>.
- [31] R. D'Sa, D. Jensen, N. Papanikolopoulos, SUAV-q - a hybrid approach to solar-powered flight, in: 2016 IEEE International Conference on Robotics and Automation (ICRA), Stockholm, Sweden, IEEE, 2016, pp. 3288–3294, <https://doi.org/10.1109/ICRA.2016.7487501>.
- [32] J. Liang, J. Buzzatto, M. Liarokapis, A tailsitter UAV based on bioinspired, tendon-driven, shape-morphing wings with aerofoil-shaped artificial feathers, in: 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, IEEE, 2023, pp. 4750–4756, <https://doi.org/10.1109/IROS55552.2023.10341538>.
- [33] V. Beliautou, A. Beliautou, Prop-plane — new convertible VTOL UAV as a combination of a longitudinal bi-copter and a flying wing with a tilt-rotor powertrain, *Aerosp. Sci. Technol.* 155 (2024) 109650, <https://doi.org/10.1016/j.ast.2024.109650>.
- [34] E.J.J. Smeur, M. Bronz, G.C.H.E. De Croon, Incremental control and guidance of hybrid aircraft applied to a tailsitter unmanned air vehicle, *J. Guid. Control Dyn.* 43 (2) (2020) 274–287, <https://doi.org/10.2514/1.G004520>.
- [35] Z. Ma, E.J.J. Smeur, G.C.H.E. De Croon, Wind tunnel tests of a wing at all angles of attack, *Int. J. Micro Air Veh.* 14 (2022) 1–11, <https://doi.org/10.1177/17568293221110931>.
- [36] H.-V. Phan, D. Floreano, A twist of the tail in turning maneuvers of bird-inspired drones, *Sci. Robot.* 9 (96) (2024) eado3890, <https://doi.org/10.1126/scirobotics.ado3890>.
- [37] E. Ajanic, M. Feroskhan, S. Mintchev, F. Noca, D. Floreano, Bioinspired wing and tail morphing extends drone flight capabilities, *Sci. Robot.* 5 (47) (2020), <https://doi.org/10.1126/scirobotics.abc2897>.
- [38] J. Zhang, Y. Liu, L. Gao, B. Liu, Y. Zhu, X. Zang, J. Zhao, H. Cai, Bioinspired drone actuated using wing and aileron motion for extended flight capabilities, *IEEE Robot. Autom. Lett.* 7 (4) (2022) 11197–11204, <https://doi.org/10.1109/LRA.2022.3192803>.
- [39] V. Wüest, S. Jeger, M. Feroskhan, E. Ajanic, F. Bergonti, D. Floreano, Agile perching maneuvers in birds and morphing-wing drones, *Nat. Commun.* 15 (2024) 8330, <https://doi.org/10.1038/s41467-024-52369-4>.
- [40] E. Ajanic, M. Feroskhan, V. Wüest, D. Floreano, Sharp turning maneuvers with avian-inspired wing and tail morphing, *Commun. Eng.* 1 (2022) 34, <https://doi.org/10.1038/s44172-022-00035-2>.
- [41] H. Huang, W. He, J. Wang, L. Zhang, Q. Fu, An all servo-driven bird-like flapping-wing aerial robot capable of autonomous flight, *IEEE/ASME Trans. Mechatron.* 27 (6) (2022) 5484–5494, <https://doi.org/10.1109/TMECH.2022.3182418>.
- [42] H. Huang, W. He, Q. Fu, X. He, C. Sun, A bio-inspired flapping-wing robot with cambered wings and its application in autonomous airdrop, *IEEE/CAA J. Autom. Sinica* 9 (12) (2022) 2138–2150, <https://doi.org/10.1109/JAS.2022.106040>.
- [43] T. Mao, B. Duan, B. Yin, C. Guo, Motion attitude and aerodynamic characteristics research of flapping wings driven by micro servomotor, *J. Bionic Eng.* 21 (2024) 2830–2846, <https://doi.org/10.1007/s42235-024-00598-0>.
- [44] L. Chen, C. Cheng, C. Zhou, Y. Zhang, J. Wu, Flapping rotary wing: a novel low-reynolds number layout merging bionic features into micro rotors, *Prog. Aerosp. Sci.* 146 (2024) 100984, <https://doi.org/10.1016/j.paerosci.2024.100984>.
- [45] D. Ma, B. Song, S. Gao, D. Xue, J. Xuan, Designing efficient bird-like flapping-wing aerial vehicles: insights from aviation perspective, *Bioinspir. Biomim.* 19 (6) (2024) 061001, <https://doi.org/10.1088/1748-3190/ad88c4>.
- [46] S. Chen, L. Yuan, J. Xiang, S. Guo, Y. Xie, K. Lan, Y. Cheng, Data-driven surrogate aerodynamic prediction model and lift optimization for the flapping wing rotor, *Aerosp. Sci. Technol.* 168 (2026) 110715, <https://doi.org/10.1016/j.ast.2025.110715>.
- [47] Y. Pan, S. Guo, J. Whidborne, X. Huang, Aerodynamic performance of a flyable flapping wing rotor with dragonfly-like flexible wings, *Aerosp. Sci. Technol.* 148 (2024) 109090, <https://doi.org/10.1016/j.ast.2024.109090>.
- [48] Z. Wei, J. Shi, Y. Wang, Y. Lyu, Y. Fu, X. Wang, Fixed-time attitude control of flying-wing unmanned aerial vehicles based on plasma control surface, *Aerosp. Sci. Technol.* 159 (2025) 110016, <https://doi.org/10.1016/j.ast.2025.110016>.
- [49] S.P. Sane, M.H. Dickinson, The control of flight force by a flapping wing: lift and drag production, *J. Exp. Biol.* 204 (15) (2001) 2607–2626, <https://doi.org/10.1242/jeb.204.15.2607>.
- [50] M.H. Dickinson, F.-O. Lehmann, S.P. Sane, Wing rotation and the aerodynamic basis of insect flight, *Science* 284 (5422) (1999) 1954–1960, <https://doi.org/10.1126/science.284.5422.1954>.
- [51] H.E. Taha, M.R. Hajj, P.S. Beran, State-space representation of the unsteady aerodynamics of flapping flight, *Aerosp. Sci. Technol.* 34 (2014) 1–11, <https://doi.org/10.1016/j.ast.2014.01.011>.
- [52] R. Schwab, J. Reade, M. Jankauski, Quasi three-dimensional deformable blade element and unsteady vortex lattice reduced-order modeling of fluid–structure interaction in flapping wings, *Phys. Fluids* 34 (2022) 121903, <https://doi.org/10.1063/5.0129128>.
- [53] D.D. Chin, D. Lentink, Flapping wing aerodynamics: from insects to vertebrates, *J. Exp. Biol.* 219 (7) (2016) 920–932, <https://doi.org/10.1242/jeb.042317>.
- [54] T. Mao, C. Guo, B. Duan, An improved quasi-steady model capable of calculating flexible deformation for bird-sized flapping wings, *Nonlinear Dyn.* 113 (2025) 5591–5610, <https://doi.org/10.1007/s11071-024-10570-6>.
- [55] S. Shkarayev, D. Silin, Measurements of aerodynamic coefficients for flapping wings at 0–90 angles of attack, *AIAA J.* 50 (10) (2012) 2034–2042, <https://doi.org/10.2514/1.J051051>.
- [56] T. Mao, B. Duan, C. Guo, Robust attitude controller design for bi-copter on SO(3), *Int. J. Control Autom. Syst.* 23 (2025) 3100–3114, <https://doi.org/10.1007/s12555-025-0226-z>.
- [57] S.P. Sane, M.H. Dickinson, The aerodynamic effects of wing rotation and a revised quasi-steady model of flapping flight, *J. Exp. Biol.* 205 (8) (2002) 1087–1096, <https://doi.org/10.1242/jeb.205.8.1087>.
- [58] M.R.A. Nabawy, W.J. Crowther, On the quasi-steady aerodynamics of normal hovering flight part II: model implementation and evaluation, *J. R. Soc. Interface* 11 (94) (2014) 20131197, <https://doi.org/10.1098/rsif.2013.1197>.
- [59] M. Okamoto, K. Yasuda, A. Azuma, Aerodynamic characteristics of the wings and body of a dragonfly, *J. Exp. Biol.* 199 (2) (1996) 281–294, <https://doi.org/10.1242/jeb.199.2.281>.
- [60] R.T. Jones, Correction of the Lifting-Line Theory For the Effect of the chord, NACA Technical Note 817, National Advisory Committee for Aeronautics, Washington, DC, 1941, <https://ntrs.nasa.gov/citations/19930085047>.
- [61] M.R.A. Nabawy, W.J. Crowther, On the quasi-steady aerodynamics of normal hovering flight part I: the induced power factor, *J. R. Soc. Interface* 11 (93) (2014) 20131196, <https://doi.org/10.1098/rsif.2013.1196>.
- [62] X. He, Y. Wang, Design and trajectory tracking control of a new bi-copter UAV, *IEEE Robot. Autom. Lett.* 7 (4) (2022) 9191–9198, <https://doi.org/10.1109/LRA.2022.3190090>.
- [63] Y. Li, Y. Qin, W. Xu, F. Zhang, Modeling, identification, and control of non-minimum phase dynamics of bi-copter UAVs, in: 2020 IEEE/ASME International Conference on Advanced Intelligent Mechanisms (AIM), 2020, pp. 1249–1255, <https://doi.org/10.1109/AIM43001.2020.9158910>.
- [64] E. Chang, D.D. Chin, D. Lentink, Bird-inspired reflexive morphing enables rudderless flight, *Sci. Robot.* 9 (96) (2024) eado4535, <https://doi.org/10.1126/scirobotics.ado4535>.
- [65] Y. Yu, S. Yang, M. Wang, C. Li, Z. Li, High performance full attitude control of a quadrotor on SO(3), in: 2015 IEEE International Conference on Robotics and Automation (ICRA), 2015, pp. 1698–1703, <https://doi.org/10.1109/ICRA.2015.7139416>.